

Unified Theory Based on Time-Wave Dynamics (The P-model)

Part I: Theoretical Foundations and Mapping Principles

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Abstract

This paper (Part I) fixes only three items as the **reference baseline** of the P-model. (i) **Mapping**: from the scalar limit of time-wave density, $P(x)$, and its covariant extension $P_\mu = (P_t, P_1, P_2, P_3)$, we define the potential ϕ and fix the main equations that map it to the observables of gravity, clocks, light propagation, and rotational effects, including the sign convention. (ii) **Freezing**: the light-propagation parameter β is not fitted ad hoc but derived as the theoretical prediction $\beta = 1$ from the null geodesics of the effective metric (1A.1), and is managed in three layers together with its operational frozen value and terminal audit status (3.4). Parts II and III do not change this value locally; if an update becomes necessary, Part I is revised and all downstream outputs are regenerated from here. (iii) **Falsification conditions**: for each observable treated in Parts II and III, we define a machine-readable falsification pack that fixes, under this mapping, which statistic must exceed which threshold for rejection. In addition, the frozen action skeleton defined here provides the theoretical anchor points on the quantum side that are audited in Parts III-A/B: the transverse photon branch, the selector-law anchor of the fine-structure constant, and the coupled-localization anchor for the W/Z boson masses. Their derivation and acceptance criteria are governed by Parts III-A/B. In this manuscript, “closed” means **mathematical closure**, not truth of the theory or eventual observational acceptance. Individual application tests in the Solar System, dynamical phenomena, strong-field gravity, cosmology, and quantum physics are collected in Parts II and III; here they are referenced only as needed.

1 Introduction

1.1 Background

The P-model operationally defines the stationary limit $P(x)$ as the base variable of time-wave density, and describes systems with rotation or flow by its covariant extension $P_\mu = (P_t, P_1, P_2, P_3)$ (stationary-clock mapping: $(\frac{d\tau}{dt})_{v=0} = \frac{P_\infty}{P}$). The aim of this paper is not to make ontological declarations or engage in philosophical dispute, but to offer a **testable hypothesis**—a unified mapping—for three unresolved questions: (i) what time is, (ii) why gravity appears, and (iii) how quantum correlations should be understood.

The minimum assumption adopted here is that matter is treated as a bound wave, light as a free wave, and the rate of a standard clock is controlled by the stationary-limit field $P(x)$ (and by the extension P_μ when rotation and flow are included). What matters is not to identify the microscopic ontology of P first, but to **fix the mapping** by which spatial changes in the ratio P/P_∞ appear in gravity, clocks, and light propagation, and to state that mapping in a form that can be rejected by primary data through explicit observables and thresholds.

Part I is the **reference document** that fixes the P-model's symbol conventions, minimum assumptions, and mappings from theory to observables (gravity, clocks, and light). The later Part II (astrophysics and cosmology) and Part III (quantum physics) follow the conventions and frozen parameters fixed here, and use them to perform primary-data-based validation and define falsification conditions. Accordingly, this manuscript minimizes the listing of individual empirical results and instead clarifies the definitions, mappings, and frozen values that the later parts must inherit.

The practical goal of the project is to test, in a reproducible way, whether the same weak-field observables can be recovered within a **scalar representation that is easier to explain physically and operationally**.

P-model viewpoint (the axis of this paper): (i) gravity is defined as the spatial gradient of clock rate, namely the gradient of P ; (ii) bound waves (clocks) and free waves (propagation modes) respond to the same P through the null-geodesic structure of the effective metric, yielding $\beta = 1$ as a theoretical consequence (1A.1); and (iii) cosmological redshift is treated not as geometry but as a mapping of the background P field. These points are not obtained by reinterpreting existing theories from within themselves. They are fixed as a new mapping and then tested against primary data in a falsifiable form.

Differential predictions that remain after freezing: once the mappings and frozen parameters are fixed, the main empirical decision points are (1) the terminal status of the free-wave propagation coefficient β through the LLR, MESSENGER, and VLBI eligibility chain, (2) strong-gravity shape observables tied to the nonlinear structure of the P gradient, (3) distance-indicator-independent cosmological diagnostics that test the background- P mapping, and (4) the procedural dependence of quantum interference correlations together with the description of microscopic interaction structure (spin and chirality) under the P_μ extension.

Core definition (Part I): P is operationally defined as the quantity that sets the rate of a standard clock. This paper fixes the mappings from P to clocks, motion, and light, and validates them only in a form that can be rejected by primary data. Reference frameworks (existing theories) are used only for comparison of final observable expressions and are not allowed to redefine the P-model internally.

Reference note: General relativity (GR) and special relativity (SR) have been extraordinarily successful in classical tests, but tracing exactly which assumptions enter an observable and its validation pipeline requires a formal framework, for example PPN parameteri-

zation.[2] Quantum mechanics (QM) is also foundational to modern physics, yet the status of time and its unification with gravity remain open questions. The P-model is positioned as a framework that re-derives the same observables from a different mechanism—the spatial variation of time-wave density P —and does not replace its definitions by importing concepts from the reference frameworks.

1.2 Contributions of This Study

- We fix the symbols and sign conventions so that the three statements “ P increases near mass,” “gravity = sliding down the P gradient,” and “light also moves toward higher P = refraction” are mathematically consistent.
- We incorporate the minimum coupling between $P_\mu = (P_t, P_1, P_2, P_3)$ and the matter current J^μ into the Part I baseline mapping, and state explicitly the exact-reduction condition $J^i = 0 \Rightarrow P_i = 0$ back to the existing scalar theory.
- We organize the pipeline from data acquisition to analysis to fixed artifacts so that a third party can reproduce the same outputs.
- We provide both a figure catalog and summary reports, making the global view and the reproducibility entry points explicit.

1.3 Scope and Claim Boundary of This Paper (Effective Model and Falsification Conditions)

This paper does not redefine the P-model by using the concepts of the reference frameworks (existing theories). Instead, it first fixes the P-model’s own minimum assumptions and mappings. The reference frameworks are used only for comparing final expressions of results, namely for consistency checks and comparisons of predictive accuracy.

From this point onward, the subject is always the P-model. The mappings, freezing rules, and rejection conditions fixed in the main text are not reinterpreted or modified in terms of the reference frameworks.

Definition of an effective model: in this paper, an “effective model” is a framework in which the action principle (Lagrangian) is defined in minimal form, the mapping to observables is fixed first, and consistency with primary data is then tested under that mapping. It is therefore distinct from pure phenomenology because it can present falsifiable predictions, namely explicit rejection conditions.

Definition of “closed”: in this paper, “closed” means mathematical closure in the sense that the P-model side has already supplied the action skeleton, interaction dictionary, mappings to observables, and falsification conditions across the major domains of modern physics. This does not mean that the theory is true or that it will eventually be accepted by future observations. P-model v2.0 is the version that fixes this mathematical closure and leaves its empirical acceptance or rejection to public primary data and future observations.

Items called “unresolved” in this paper are not of a single kind. They are classified into the following four layers.

- **Theoretically closed:** the P-model’s mathematical dictionary and anchor points are already fixed.
- **Adopted effective sector:** the derivation of the theoretical origin remains open, but the observational interface is already fixed (for example, retaining local $U(1)$).

- **Bridge residual:** a remaining task in the mapping to observables, such as four-dimensional shape factors or operator corrections.
- **Awaiting observational decision:** an item that must wait for public primary data or future observations.

All remaining issues from this point onward belong to one of these four layers; they are not treated as missing new laws of physics.

This "effective model" is read in two phases. In the theoretical phase, the P-model provides the reference surface of an action skeleton and observable mappings that mathematically closes the major domains of modern physics. In the observational phase, it is operated as an effective theory whose reference surface is tested against primary data by falsification-condition packs.

The main concern of this paper is to test, with primary sources and reproducible outputs, how consistently the observables of gravity, clocks, light propagation, and rotational effects can be described once the stationary-limit time-wave density $P(x)$ and its covariant extension P_μ are introduced. For that reason, this paper does not claim to present the ontological completion of P. Here "completion" would mean identifying the microscopic entity behind P, or presenting a final formalism in which all interactions are derived from a single principle. The mathematical closure of v2.0 stands independently of that layer. The remaining tasks instead belong to observational audit, numerical constraints, and the fixing of bridge quantities. Meanwhile, the reference action skeleton is already defined in Part I 2.8, and the acceptance of derivational closure and drift is fixed mechanically by the audit in Part IV, Section 3. In Part I, the static minimal equation in its weak-field limit (of Poisson type) is adopted, and the work is positioned as an **effective model** in which the mapping to observables is fixed first and then tested. Accordingly, P is not first declared to be some particular underlying substance. It is treated only as a mapping that appears in observables through the ratio P/P_∞ .

The parameter β is managed in three layers: the theoretical prediction ($\beta = 1$), the operational frozen value ($\beta = 1.0000105$), and the terminal audit status (currently Pass). This paper does not optimize that value case by case in order to obtain results. Instead, it is fixed as defined in Part I and then used for extrapolation (prediction). This paper does not claim theoretical superiority over the reference frameworks in the sense of first-principles reducibility.

The following operating rules are stated explicitly to avoid misunderstanding.

- **The responsibility for freezing is fixed.** The theoretical prediction, operational frozen value, and terminal audit status of β are defined in this paper (Part I 3.4). The texts of Parts II and III do not alter the value locally. If an update is required, it is consolidated into a revision of Part I, after which all downstream comparisons, figures, and falsification conditions are regenerated together.
- **Fitting and prediction are separated.** (Note) fitting is restricted to the procedure that determines the frozen values; no further optimization is performed in producing later results. β is frozen from primary-source constraints and is thereafter used only for extrapolation (prediction), with the frozen values collected in the frozen-parameter set.
- In the weak-field regime (the Solar System), agreement with the reference frameworks at high precision is required, while in strong fields, dynamical phenomena, and cosmology the differential predictions are expected to appear in observable form (the concrete examples and required precision are collected in Parts II and III).
- Some public primary sources (geodetic systems, time systems, catalogs, and so on) may include preprocessing based on the standard theory. In this paper, priority is given first to fixing the pri-

mary source inputs and visualizing corrections and systematics through fixed outputs; discussion of raw-data reanalysis is treated separately when necessary.

- Falsifiability is emphasized by fixing the falsification-condition packs, namely which observable and which threshold must be exceeded for rejection, and not moving them afterward in an ad hoc way.
- The connection to quantum physics is positioned as a validation target of Part III, while this paper (Part I) fixes only the minimal dictionary required for that connection (Section 2.8).

The limitations of this paper at the present stage are as follows.

- The action skeleton is already defined, and the acceptance or rejection of derivational closure is determined mechanically by the audit in Part IV, Section 3. At the same time, more highly refined dynamics for strong fields and many-body systems remain in progress. This does not negate the mathematical closure of v2.0; it belongs to the tracking of higher-order effects under the frozen action skeleton.
- The microscopic origin of β lies outside the scope of this paper, and at the current stage β is frozen from primary-source constraints and then tested. The same is true for extension hypotheses such as velocity saturation; they are not part of the core mapping of this paper.
- The connection to quantum phenomena remains at the operational level (its placement is treated in Part III).
- The essential conditions of the covariant extension P_μ (dipole-radiation prohibition, quadrupole leading term, and the exact stationary-limit reduction $J^i = 0 \Rightarrow P_i = 0$) are already fixed by definition, and their acceptance is continuously audited by the same interface in Parts II and IV.

These are present-stage constraints of the P-model and should be resolved by future theoretical development.

The list of primary sources (URLs, access dates, and local storage policy) and the reproducibility entry points (fixed artifacts, reproduction commands, and audit gates) are collected in Part IV (Verification Materials) (GitHub: [EnterOgawa/p-model](#)) and are kept separate from the main body of the paper. In particular, the acceptance of action-principle derivational closure is governed by Part IV, Section 3, and the fixed grounds for the Part I core mapping are governed by Part IV, Section 10.[1]

Reference note: this paper does not adopt the reference frameworks (existing theories) as its foundation. It fixes the P-model side mappings and frozen values first, and limits the reference frameworks to comparison of final expressions.

Style Rules The main text closes using only the P-model’s own definitional mapping procedure. Comparisons with GR, QM, PPN, and so forth are isolated in reference notes at the end of sections and are not allowed to replace definitions or derivations in the main text. In this paper, a reference note is defined as a non-normative note that records only comparison with, and placement relative to, the reference frameworks. It is not used for the main text’s definitions, derivations, or decision conditions.

2 Theory

The main symbols and sign conventions are collected in the appendix (Symbols and Definitions), and only the necessary items are repeated in the main text.

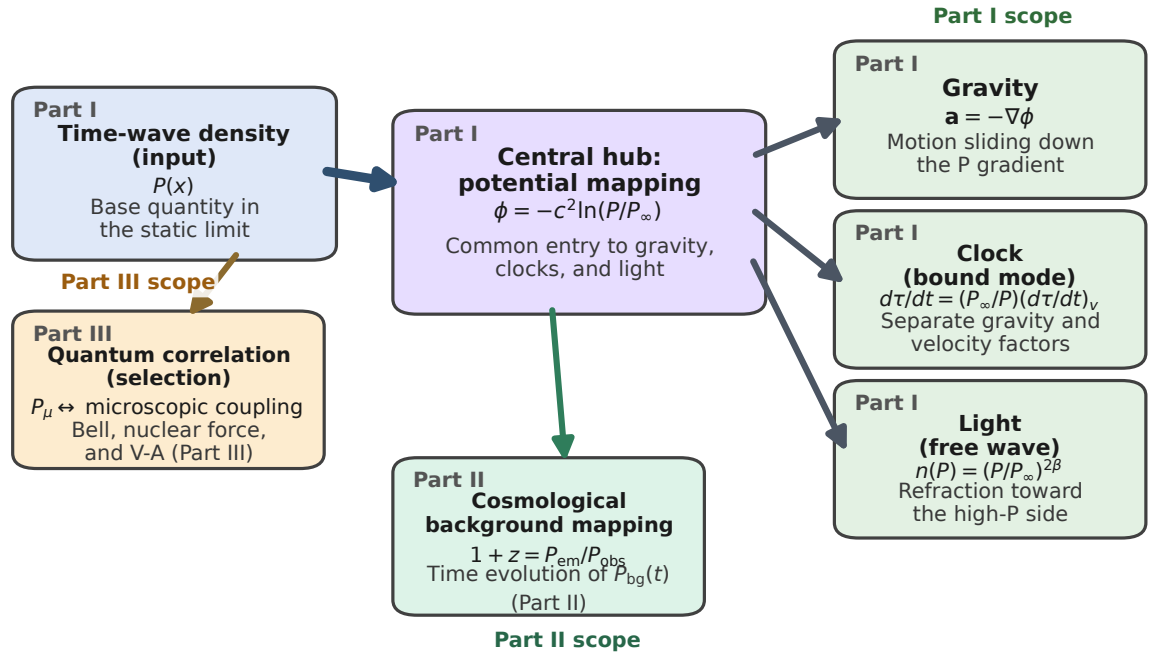
2.1 Unified Interpretation (Written in P)

The central claim of the P-model is that gravity, clocks, and light propagation can be expressed consistently through the stationary-limit time-wave density $P(x)$ and its extension to P_μ for systems with rotation and flow. In this paper, matter is regarded as a bound wave and light as a free wave. The "shortest path" followed by a wave is interpreted not as geometric distance but as the stationarity condition of the **arrival time (optical path length)**.

Under that interpretation, the mapping to observables is unified in the following form (details are given in Sections 2.1-2.8):

- gravitational potential: $\phi = -c^2 \ln(P/P_\infty)$
- gravity (equation of motion): $\mathbf{a} = -\nabla\phi = c^2 \nabla \ln(P/P_\infty)$
- clock rate: $\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt}\right)_v$
- light: $n(P) = (P/P_\infty)^{2\beta}$ ($\beta > 0$), so refraction is toward the higher-P side
- rotation and flow: by introducing the four-vector extension and the minimal coupling, the stationary-limit reduction is fixed by the following expression

$$P_\mu = (P_t, P_1, P_2, P_3), \quad \mathcal{L}_{\text{int}} = g_P P_\mu J^\mu, \quad J^i = 0 \Rightarrow P_i = 0, \quad P_t \equiv P \quad (1)$$



Part I fixes the mapping and β_{frozen} ; Parts II/III run falsification audits with those frozen values.

Figure 1: Core mapping architecture of the P-model. With ϕ as the central hub, the mappings to each observable channel are arranged radially, and the jurisdictions of Parts I/II/III are color-coded.

Table (conceptual comparison):

Viewpoint	P-model (this paper)	Reference framework (GR)
gravity	$\phi = -c^2 \ln(P/P_\infty)$	curvature of the metric field (displayed as an effective potential in the weak field)
motion	$\mathbf{a} = -\nabla\phi = c^2\nabla\ln(P/P_\infty)$	weak-field, low-velocity limit of the geodesic equation
light propagation	refraction through $n(P) = (P/P_\infty)^{2\beta}$	description as a null geodesic (light ray)
clock	$\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt}\right)_v$	metric description of the proper time $d\tau$
redshift	time mapping of the background wave $P_{\text{bg}}(t)$	description by the FRW scale factor $a(t)$

In this paper, GR is treated as a reference framework, and the mappings to observables are fixed by the P-model-side definitions (the left column of the table above).

2.2 P and the Potential ϕ

$P(x)$ denotes the **stationary-limit time-wave density** at the spatial point x , and P_∞ denotes the far-field reference density. For systems with rotation and flow, the covariant extension $P_\mu = (P_t, P_1, P_2, P_3)$ is used (Section 2.8). In this paper, P_∞ is used strictly as the far-field reference, while P_t is used as the time component of the four-vector. P_∞ is a boundary condition for the **local systems (the asymptotically flat region from Solar-System to galactic scales)** considered in this paper (Part I), and is treated as constant over the timescales of experiments and observations. In cosmology, $P_{\text{bg}}(t)$ is introduced and the decomposition $P = P_{\text{bg}}(t)P_{\text{local}}(x)$ is used; the time evolution of the background is treated in Part II. This paper (Part I) does not identify the microscopic origin of P (for example, whether it is an energy density, information, or something else). The connection to the quantum vacuum and to local P structure is deferred to Part III. Instead, this paper adopts an **operational definition**. Namely, P is defined as the stationary-limit scalar component of the field that determines the rate of a standard clock, and the stationary-clock mapping $((\frac{d\tau}{dt})_{v=0} = \frac{P_\infty}{P})$

$$\left(\frac{d\tau}{dt}\right)_{v=0} = \frac{P_\infty}{P} \quad (2)$$

is taken as the observational definition of P (Section 2.3). What appears in observables under this mapping is only the **ratio** P/P_∞ (**dimensionless**). Accordingly, in the stationary-limit channels the theory closes mathematically if P is treated as a dimensionless scalar field (or equivalently as a quantity of the same dimension as P_∞). To define $\ln(P/P_\infty)$, P and P_∞ must have the same dimension. The rotation/flow channels are treated by extending to P_μ in Section 2.8. In this project, the convention $P > P_\infty$ **near mass** is adopted.

This paper takes the basic quantity not to be the absolute value of P but the **dimensionless ratio** P/P_∞ , and further adopts $u(x) \equiv \ln(P/P_\infty)$ as the fundamental variable on the potential side.

$$u(x) \equiv \ln\left(\frac{P}{P_\infty}\right) \quad (3)$$

Using this u , the Newtonian-form potential (zero at infinity, negative near mass) is defined by

$$\phi \equiv -c^2 \ln\left(\frac{P}{P_\infty}\right) \quad (4)$$

that is, $\phi = -c^2 u$.

Requirement (internal to P) $\rightarrow$ consequence: what appears in observations is not the absolute magnitude but only the ratio P/P_∞ . Local delay effects (the rate of a clock) compose multiplicatively, while the potential-like effect entering the equation of motion must be handled as an additive quantity. Therefore, the logarithm is adopted as the minimal mapping that turns multiplicative composition into additive composition, and the expression above is fixed as a definition of the P-model.

Accordingly, ϕ has the same dimension as the Newtonian potential (the dimension of velocity squared). For a point mass M in the weak, static regime, one has $u(r) = GM/(c^2 r)$, and therefore

$$\frac{P(r)}{P_\infty} = e^{\frac{GM}{c^2 r}}, \quad \phi(r) = -\frac{GM}{r} \quad (5)$$

which recovers the Newtonian limit (presented here as a consistency check rather than as an explanation).

2.3 Gravity = Sliding Down the P Gradient

The gravitational acceleration is unified in Newtonian form as

$$\mathbf{a} = -\nabla\phi = c^2 \nabla \ln \left(\frac{P}{P_\infty} \right) \quad (6)$$

This makes the phrase "gravity = sliding down the P gradient" identical to its mathematical expression. Within the scope of this paper (Part I), namely the **static weak field**, the Poisson-type equation is adopted as the minimal form (the dynamical equation is discussed in Section 2.6). For the mass density ρ ,

$$\nabla^2\phi = 4\pi G\rho, \quad \nabla^2 \ln \left(\frac{P}{P_\infty} \right) = -\frac{4\pi G}{c^2}\rho \quad (7)$$

Applying the same weak-field source normalization to the electromagnetic derivation in Part III-A 2.6 yields the coupling ratio $g_P/Z_P = 4\pi G$.

This is adopted, and in vacuum one has $\nabla^2\phi = 0$ (equivalently, $\nabla^2 \ln(P/P_\infty) = 0$). Since many weak-field tests in the Solar System can be discussed from the properties of the vacuum solution, this is sufficient for the scope of this paper.

2.4 Clocks (Gravitational Term + Velocity Term)

Separating the observational time t (external reference time) from the proper time τ (the clock's own time), the rate of a clock is written as

$$\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt} \right)_v \quad (8)$$

For a stationary clock ($v = 0$), in the weak field

$$\frac{P_\infty}{P} = e^{\phi/c^2} \approx 1 + \frac{\phi}{c^2} \quad (9)$$

which agrees with the first-order gravitational redshift. The velocity term is

$$\left(\frac{d\tau}{dt} \right)_v = \sqrt{1 - \frac{v^2}{c^2}} \quad (10)$$

This velocity term is not an assumption borrowed from outside the model. It is derived directly from the particle-limit action fixed in Section 2.8.1,

$$S_a = -m_a c \int e^{-\chi} \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} \quad (11)$$

Writing the worldline as $x^\mu = (ct, \mathbf{x}(t))$, one obtains

$$\sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} = c \sqrt{1 - \frac{v^2}{c^2}} dt \quad (12)$$

and the factor $e^{-\chi}$ in the integrand gives the gravitational term P_∞/P_t . Therefore,

$$\frac{d\tau}{dt} = e^{-\chi} \sqrt{1 - \frac{v^2}{c^2}} = \frac{P_\infty}{P_t} \sqrt{1 - \frac{v^2}{c^2}} \quad (13)$$

is obtained. On the stationary branch, $P_t = P$, so the clock formula used in this section,

$$\frac{d\tau}{dt} = \frac{P_\infty}{P} \sqrt{1 - \frac{v^2}{c^2}} \quad (14)$$

is recovered exactly.

Reference note: the velocity term in this section has the same form as the time dilation of SR (special relativity) in the reference framework. Experimental comparisons involving clock systems (gravitational redshift and satellite clocks) are treated in Part II.[3][6][7]

2.5 Light (Refraction: Bending Toward Higher P)

Requirement (response of the free wave): bound waves (clocks) and free waves (propagating waves) may have different responses to the same P. In this section, the response of the free wave is expressed with the minimal degree of freedom, namely a single exponent. The refractive index is therefore fixed as the one-parameter family $n(P) = (P/P_\infty)^{2\beta}$, and β is treated as a quantity frozen by primary data. Light (the free wave) is assumed to have a refractive index that changes with P:

$$n(P) = \left(\frac{P}{P_\infty}\right)^{2\beta} \quad (\beta > 0) \quad (15)$$

This form is adopted. The light path is determined by Fermat's principle, namely the stationarity of the optical path length $\int n ds$, and therefore bends toward higher P. Because $\beta > 0$ is adopted here, near mass ($P > P_\infty$) one has $n > 1$, so $c_{\text{eff}}(P) = c/n(P)$ decreases. Hence, optically, light bends toward the higher-P side (the higher- n side).

Also, when this paper says that "a wave prefers the shortest path," it does not mean minimization of the geometric distance $\int ds$. It means the stationarity condition of the optical path length $\int n ds$, equivalently of arrival time or phase accumulation.

Important (to avoid confusion): the factor 2 that appears on the light-propagation side is carried by β . Therefore the definition of ϕ in Section 2.1 is fixed, and no square root is inserted on the ϕ side to absorb that coefficient.

Reference note: if PPN is used as the reference framework, the coefficient for light can be compared by the correspondence $1 + \gamma = 2\beta$ (the factor 2 is carried on the β side). The treatment of weak-field primary constraints (for example Cassini) and independent cross-checks (for example VLBI), together with the strong-field differential predictions, is collected in Part II.[2][5][4]

2.6 Cosmology (Redshift from the Background P: Fixing the Mapping)

On cosmological scales, the local-gravity contribution $P_{\text{local}}(x)$ is separated from $P_{\text{bg}}(t)$, which provides the time standard of the universe as a whole, so that

$$P(x, t) = P_{\text{bg}}(t) P_{\text{local}}(x) \quad (16)$$

This represents the time variation of the background P. If the observed frequency ratio is written as $\nu_{\text{obs}}/\nu_{\text{em}} = P_{\text{obs}}/P_{\text{em}}$, then the redshift is given by

$$1 + z = \frac{P_{\text{em}}}{P_{\text{obs}}} \quad (17)$$

Thus redshift can be expressed using only the time variation of $P_{\text{bg}}(t)$, without assuming spatial expansion. This paper (Part I) only fixes this observational mapping (background P $\rightarrow$ redshift and distance indicators). The validation and rejection decisions involving distance indicators, first-order statistics (BAO), and JWST primary-spectrum entry points are collected in Part II (Astrophysics).

Reference note: for late-universe comparison, see the standard parameter summary of Planck 2018 and the primary analysis of BOSS DR12 BAO.[13][14]

2.6.1 Distance Indicators (D_L , D_A) and the Minimal Distance Duality of the P-model

Even without assuming spatial expansion, the **definitions** of the distance indicators used in observation can be adopted unchanged:

- luminosity distance: $F_{\text{obs}} \equiv L_{\text{em}}/(4\pi D_L^2)$
- angular-diameter distance: $D_A^2 \equiv dA_{\text{em}}/d\Omega_{\text{obs}}$

In the minimal P-model (static geometry + background-P redshift), time stretching is not introduced as an independent assumption. It is derived from the phase interval of a continuous wave. If the radiating source marks the same phase difference 2π every period Δt_{em} , then the same phase difference is preserved on the observer side, and therefore

$$\nu_{\text{em}}\Delta t_{\text{em}} = \nu_{\text{obs}}\Delta t_{\text{obs}} \quad (18)$$

holds. Here the frequency ratio is given by the background-P mapping

$$\frac{\nu_{\text{obs}}}{\nu_{\text{em}}} = \frac{P_{\text{obs}}}{P_{\text{em}}} = \frac{1}{1+z} \quad (19)$$

and therefore

$$\Delta t_{\text{obs}} = (1+z)\Delta t_{\text{em}} \quad (20)$$

is obtained. Below, this time-stretch relation is called Condition A. In observational shorthand, it corresponds to $p_t = 1$. Further, using photon-number conservation (no opacity) together with $E \propto \nu$, the flux becomes

$$F_{\text{obs}} = \frac{L_{\text{em}}}{4\pi D^2(1+z)^2} \quad (21)$$

where the $(1+z)^2$ factor is the attenuation from "photon energy x arrival rate." Therefore, by definition,

$$D_L^{(P)} = (1+z)D, \quad D_A^{(P)} = D \quad (22)$$

is obtained, and the minimal distance duality is

$$D_L^{(P)} = (1+z)D_A^{(P)} \quad (23)$$

From here, this paper defines the P-model-specific DDR diagnostic as

$$\eta^{(P)}(z) \equiv \frac{D_L}{(1+z)D_A} \quad (24)$$

and the minimal P-model requires $\eta^{(P)} = 1$. For comparison, in FRW one has $D_L = (1+z)^2 D_A$, so inserting the observational pair (D_L, D_A) into the same diagnostic formula gives $\eta^{(P)} = (1+z)$. Thus, this is not the operation "substitute FRW into the P-model formula"; it is the operation of **comparing both hypotheses with the same diagnostic quantity**. This contrast, **constant vs. $(1+z)$** , is the operational discriminant.

The relation above can be modified if opacity (α), evolution of standard candles/rulers (s_L , s_R), or the

time stretch derived from wave phase (Condition A) fails. The detailed derivation and the validation against public primary sources, including the dependence on those assumptions, are collected in Part II (with the fixed registry gathered in the verification materials).[1]

2.7 Dynamical P (Radiation and Waves)

Sections 2.1-2.4 dealt mainly with gravity, clocks, and light propagation under the assumption of a **static time-wave density** $P(x)$. To handle phenomena that include "radiation," such as binary orbital decay and gravitational-wave chirps, one needs the dynamics of $P(x, t)$.

In this paper (Part I), the dynamical P sector is fixed only up to the **minimal derivation (the retarded solution of a uniformly moving point source)**, while the observational validation is deferred to Part II (Astrophysics).

- **Basic variable:** use $u(x, t) \equiv \ln(P/P_\infty)$ (that is, $u = -\phi/c^2$).
- **Minimal dynamical assumption:** in order to connect to the Newtonian limit in the static regime, adopt in the weak field

$$\left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2\right) u = -\frac{4\pi G}{c^2} \rho \quad (25)$$

as the minimal wave equation for the weak field (in vacuum, $\square u = 0$).

- **Derivation of the retarded potential (uniformly moving point source):** rewrite the above in the equivalent form

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2\right) u = \frac{4\pi G}{c^2} \rho \quad (26)$$

and substitute a point source moving at constant velocity,

$$\rho(\mathbf{r}, t) = M \delta^3(\mathbf{r} - \mathbf{v}t) \quad (27)$$

Using the retarded Green's function,

$$u(\mathbf{r}, t) = \frac{G}{c^2} \int d^3\mathbf{r}' \frac{\rho(\mathbf{r}', t_r)}{|\mathbf{r} - \mathbf{r}'|}, \quad t_r = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c} \quad (28)$$

one obtains, for a point source,

$$u(\mathbf{r}, t) = \frac{GM/c^2}{(1 - \mathbf{n}_r \cdot \boldsymbol{\beta}_v) R_r} \Big|_{t_r} \quad (29)$$

where $R_r = |\mathbf{r} - \mathbf{r}_s(t_r)|$, $\mathbf{n}_r = R_r^{-1}(\mathbf{r} - \mathbf{r}_s(t_r))$, and $\boldsymbol{\beta}_v = \mathbf{v}/c$. Defining $\mathbf{s} \equiv \mathbf{r} - \mathbf{v}t$ and decomposing it as $\mathbf{s} = \mathbf{s}_\parallel + \mathbf{s}_\perp$ with $\mathbf{s}_\parallel \parallel \mathbf{v}$ gives

$$u(\mathbf{r}, t) = \frac{GM/c^2}{\sqrt{s_\parallel^2 + (1 - \beta_v^2) s_\perp^2}}, \quad \beta_v = \frac{|\mathbf{v}|}{c} \quad (30)$$

Therefore the gravitational acceleration is

$$\mathbf{a} = c^2 \nabla u = -\frac{GM \left[(1 - \beta_v^2) \mathbf{s} + \boldsymbol{\beta}_v (\boldsymbol{\beta}_v \cdot \mathbf{s}) \right]}{\left[s_\parallel^2 + (1 - \beta_v^2) s_\perp^2 \right]^{3/2}} \quad (31)$$

Since $u \propto 1/r$, the gradient ∇u points toward the source (the negative radial direction), so the leading minus sign is consistent with $\mathbf{a} = c^2 \nabla u = -\nabla \phi$ and hence $\phi = -c^2 u$. In the slow-motion limit,

$$\mathbf{a} = -GM \frac{\mathbf{s}}{|\mathbf{s}|^3} + \mathcal{O}\left(\frac{v^2}{c^2}\right) \quad (32)$$

so a **uniformly moving source does not generate an $\mathcal{O}(v/c)$ center shift**; the acceleration vector points to the current position $\mathbf{vt}$. Accordingly, any observationally significant offset Δx must be evaluated not from the retarded term of uniform motion alone, but from the full dynamical branch including acceleration, response delay, and multi-source interference (the actual audit is in Part II).

- **Radiation structure (minimum requirement):** as the protective condition that excludes dipole radiation, require an equivalence-principle-like universal coupling of matter to the time wave, namely $g_{P,a}/m_a = \kappa$ for all matter species. Then the source density of the time wave is

$$\rho_P = \kappa \rho_m \quad (33)$$

and the dipole moment is fixed as

$$D_P = \kappa M X_{\text{cm}} \quad (34)$$

In the center-of-mass frame,

$$X_{\text{cm}} = 0 \Rightarrow \ddot{D}_P = 0 \quad (35)$$

holds, so the dipole term in the far-zone radiation cancels exactly. The first nontrivial radiation that remains in the weak far field is therefore the quadrupole term, and Part II audits this condition through the recalculation of $\dot{P}_b$.

- **Fixing the derivation chain for τ and ξ :** coarse-grain the P_μ/J^μ dynamics along the collision axis and fix explicitly the intermediate steps that reduce the wave equation to a first-order delayed equation. First define the low-frequency components along the collision axis x by

$$u_L(x, t) \equiv (G_L * u)(x, t), \quad \rho_L(x, t) \equiv (G_L * \rho)(x, t) \quad (36)$$

where G_L is a coarse-graining kernel, and write the weak-field wave equation as

$$\frac{1}{c_w^2} \partial_{tt} u_L - \partial_{xx} u_L = S_L, \quad S_L \equiv \frac{4\pi G}{c^2} \rho_L \quad (37)$$

Projecting out unresolved modes with an effective equation of the Mori-Zwanzig first-memory-truncation type introduces a friction term for $\partial_t u_L$ and yields

$$\tau_{\text{free}} \partial_{tt} u_L + \partial_t u_L = D \partial_{xx} u_L + c_w^2 S_L, \quad D \equiv c_w^2 \tau_{\text{free}} \quad (38)$$

which is of telegraph type.

Next define the time-scale separation by

$$\varepsilon_{\text{slow}} \equiv \frac{\tau_{\text{free}}}{T_{\text{slow}}}, \quad T_{\text{slow}} \equiv \min(\tau_{\text{int}}, \tau_{\text{damp}}) \quad (39)$$

and in the slow-manifold regime $\varepsilon_{\text{slow}} \ll 1$, discard $\tau_{\text{free}} \partial_{tt} u_L$ to obtain

$$\partial_t u_L = D \partial_{xx} u_L + c_w^2 S_L \quad (40)$$

To return to P_L , use the linearization $P_L = P_\infty(1 + u_L) + O(u_L^2)$ valid for $|u_L| \ll 1$ together with the source target J^0 , so that the conservation law and the BGK-type relaxation

$$\partial_t P_L + \partial_x q = -\frac{P_L - J^0}{\tau_{\text{free}}} \quad (41)$$

$$q = -D \partial_x P_L - \xi[(1 - \eta)J^x + \eta J^x(t - \Delta t)] \quad (42)$$

give

$$\partial_t P_L = D \partial_{xx} P_L - \frac{P_L - J^0}{\tau_{\text{free}}} + \xi[(1 - \eta)\partial_x J^x + \eta \partial_x J^x(t - \Delta t)] \quad (43)$$

as the reduced equation.

Now define the first moment of the P -peak centroid, X_P ,

$$X_P(t) \equiv \frac{\int x P_L(x, t) dx}{\int P_L(x, t) dx} \quad (44)$$

Then the first moment of the kernel projects the dynamics to

$$\tau_{\text{eff}} \frac{dX_P}{dt} + X_P = X_g + \xi \tau_{\text{eff}} [(1 - \eta)\chi(t)v(t) + \eta R_{\text{delay}}(t)] \quad (45)$$

where

$$R_{\text{delay}}(t) = \int_0^t K_{\tau_k}(t - s) \chi(s - \Delta t) v(s - \Delta t) ds \quad (46)$$

and the characteristic times are fixed by

$$\tau_{\text{free}} = \frac{L_{\text{path}}}{c_w}, \quad \tau_{\text{int}} = \sqrt{\frac{\langle (\chi v)^2 \rangle}{\langle (d(\chi v)/dt)^2 \rangle}}, \quad \tau_{\text{damp}} = \sqrt{\frac{\langle v^2 \rangle}{\langle (dv/dt)^2 \rangle}} \quad (47)$$

$$\eta = \frac{\tau_{\text{free}}}{\tau_{\text{free}} + \tau_{\text{int}}}, \quad \tau_{\text{eff}} = \tau_{\text{free}} + \eta \tau_{\text{kernel}}, \quad \tau_{\text{kernel}} = w_f \tau_f + w_s \tau_s, \quad w_i \propto \frac{1}{\tau_i} \quad (48)$$

$$\xi = \frac{\tau_{\text{int}} + \tau_{\text{damp}}}{\tau_{\text{eff}}} \quad (49)$$

Hence τ and ξ are not externally inserted parameters; they are derived from the P_μ kernel and the J^μ transition.

- **Emergence of L_{path} and its universal scaling law (starting from $L_{\text{int}} = g_P P_\mu J^\mu$):** in the low-frequency limit obtained after eliminating the fast modes of P_μ , the correlation of J^x determines the friction rate of the P response.

$$\Gamma_{\text{path}} \equiv \tau_{\text{free}}^{-1} = \frac{g_P^2}{\chi_P} \int_0^\infty \langle \delta J_x(t) \delta J_x(0) \rangle dt \quad (50)$$

In the macroscopic branch of a collisional gas, separate the decorrelation into an advection term and a collision term:

$$\Gamma_{\text{path}} = \Gamma_{\text{adv}} + \Gamma_{\text{coll}}, \quad \Gamma_{\text{adv}} \simeq \frac{v}{L_{\text{corr}}}, \quad \Gamma_{\text{coll}} \simeq A_{\text{col}} \rho T^{-3/2} \quad (51)$$

Here A_{col} is a Coulomb-like transport coefficient, and L_{corr} is the correlation length extracted from the gradient of J^0 . The coarse-grained length scale is therefore

$$L_{\text{path}} = c_w \tau_{\text{free}} = \frac{c_w}{\frac{v}{L_{\text{corr}}} + A_{\text{col}} \rho T^{-3/2}} \quad (52)$$

and its ratio to the reference state (ρ_0, v_0, T_0) is

$$\frac{L_{\text{path}}}{L_{\text{path},0}} = \frac{1 + \Pi_0}{r_v + \Pi_0 r_\rho r_T^{-3/2}}, \quad r_\rho \equiv \frac{\rho}{\rho_0}, \quad r_v \equiv \frac{v}{v_0}, \quad r_T \equiv \frac{T}{T_0} \quad (53)$$

$$\Pi_0 \equiv \frac{A_{\text{col}} \rho_0 T_0^{-3/2}}{v_0 / L_{\text{corr}}} = \frac{\tau_{\text{int},0}}{\tau_{\text{free},0}} - 1 \quad (54)$$

so the universal predictions are

$$\begin{aligned} \rho \rightarrow 2\rho : \frac{L_{\text{path}}(2\rho)}{L_{\text{path}}(\rho)} &= \frac{1 + \Pi_0}{1 + 2\Pi_0}, \\ v \rightarrow 2v : \frac{L_{\text{path}}(2v)}{L_{\text{path}}(v)} &= \frac{1 + \Pi_0}{2 + \Pi_0}, \\ T \rightarrow 2T : \frac{L_{\text{path}}(2T)}{L_{\text{path}}(T)} &= \frac{1 + \Pi_0}{1 + \Pi_0/2^{3/2}}. \end{aligned} \quad (55)$$

Thus L_{path} becomes shorter when the density increases or the flow accelerates, while higher temperature weakens the collisional branch and lengthens L_{path} .

2.7.1 Minimal Mapping to Structure Formation ($f\sigma_8$)

To treat the density perturbation $\delta \equiv \delta\rho/\bar{\rho}$ while keeping the geometry static, linearize the continuity and Euler equations as

$$\partial_t \delta + \theta = 0, \quad \theta \equiv \nabla \cdot \mathbf{v} \quad (56)$$

$$\partial_t \theta = c^2 \nabla^2 u - c_s^2 \nabla^2 \delta \quad (57)$$

Take this as the kinematic sector. On the gravity side, use a delayed response rather than an instantaneous Poisson equation:

$$\tau_{\text{eff}} \partial_t u + u = u_N, \quad \nabla^2 u_N = -\frac{4\pi G}{c^2} \bar{\rho} \delta \quad (58)$$

Eliminating u and θ then gives, in k -space,

$$\tau_{\text{eff}} \delta^{(3)} + \delta^{(2)} + \tau_{\text{eff}} c_s^2 k^2 \delta^{(1)} + (c_s^2 k^2 - 4\pi G \bar{\rho}) \delta = 0 \quad (59)$$

where $\delta^{(n)} \equiv d^n \delta / dt^n$. In the slow-manifold regime $\varepsilon_{\text{delay}} \equiv \tau_{\text{eff}} / T_{\text{growth}} \ll 1$, this reduces to

$$\delta^{(2)} + \Gamma_{\text{eff}} \delta^{(1)} + (c_s^2 k^2 - 4\pi G \bar{\rho}) \delta = 0, \quad \Gamma_{\text{eff}} = 4\pi G \bar{\rho} \tau_{\text{eff}} \quad (60)$$

Thus, even without assuming spatial expansion, the delayed response τ_{eff} naturally generates the **effective friction term** $\Gamma_{\text{eff}} \delta^{(1)}$ that suppresses structure growth.

The mapping to observables is defined from the background field P_{bg} by

$$a_{\text{eff}}(t) \equiv \frac{P_{\text{bg}}(t_{\text{obs}})}{P_{\text{bg}}(t)} = \frac{1}{1+z}, \quad H_{\text{eff}} \equiv \frac{d \ln a_{\text{eff}}}{dt} = -\frac{d \ln P_{\text{bg}}}{dt} \quad (61)$$

Define $G_\delta \equiv \delta / \delta_{\text{ref}}$ and $x \equiv \ln a_{\text{eff}}$. Then

$$\frac{d^2 G_\delta}{dx^2} + \left(\frac{d \ln H_{\text{eff}}}{dx} + \frac{\Gamma_{\text{eff}}}{H_{\text{eff}}} \right) \frac{d G_\delta}{dx} - \left(\frac{4\pi G \bar{\rho}}{H_{\text{eff}}^2} - \frac{c_s^2 k^2}{H_{\text{eff}}^2} \right) G_\delta = 0 \quad (62)$$

is fixed as the minimal growth equation. The growth rate is then

$$f \equiv \frac{d \ln G_\delta}{d \ln a_{\text{eff}}}, \quad f \sigma_8 = f \sigma_8(a_{\text{eff}}) \quad (63)$$

which provides the quantity used in the RSD audit of $f \sigma_8(z)$ in Part II 5.3.2.

- **Theoretical limitation (strong field):** the above equation is the weak-field linearized minimum form. In strong fields, such as near black holes or just before compact-binary merger, higher-order terms may become non-negligible. Candidate corrections include self-interaction terms such as u^2 and higher-order kinetic terms such as $\partial_\mu u \partial^\mu u$. In this paper, however, priority is given to fixing the mapping, while the strong-field equation itself and its validation are left to Part II.

2.7.2 Radiation-Dominated Limit of the Background Wave P_{bg} and the Derivation of $q_B = 1/2$

For the weak-field scalar branch of the action skeleton fixed in Part I, take the equation of motion for $u \equiv \ln(P/P_\infty)$ to be

$$\frac{1}{c^2} \partial_{tt} u - \nabla^2 u = \frac{4\pi G}{c^2} \rho_{\text{eff}}, \quad \rho_{\text{eff}} \equiv \rho + \frac{3p}{c^2} \quad (64)$$

For the cosmological background $u_{\text{bg}}(t)$, the field is spatially uniform so that $\nabla^2 u_{\text{bg}} = 0$, giving

$$\ddot{u}_{\text{bg}} = 4\pi G \left(\rho + \frac{3p}{c^2} \right) \quad (65)$$

Using early-universe blackbody radiation as the source, together with the equation of state $p = \mathcal{U}_{\text{rad}}/3$ and $\rho = \mathcal{U}_{\text{rad}}/c^2$, this becomes

$$\rho_{\text{eff}} = \frac{2\mathcal{U}_{\text{rad}}}{c^2}, \quad \ddot{u}_{\text{bg}} = \frac{8\pi G}{c^2} \mathcal{U}_{\text{rad}}. \quad (66)$$

Now use the blackbody relation $\mathcal{U}_{\text{rad}} = a_{\text{rad}} T^4$ together with the background mapping $T \propto P_{\text{bg}}$, that is, $T = T_B e^{u_{\text{bg}} - u_B}$:

$$\mathcal{U}_{\text{rad}} = \mathcal{U}_B^{(\text{rad})} e^{4(u_{\text{bg}} - u_B)}, \quad \ddot{u}_{\text{bg}} = \Omega_B^2 e^{4(u_{\text{bg}} - u_B)}, \quad \Omega_B^2 \equiv \frac{8\pi G}{c^2} \mathcal{U}_B^{(\text{rad})}. \quad (67)$$

Assume the asymptotic form in the early-time limit $t \rightarrow 0^+$,

$$u_{\text{bg}}(t) = u_B - q_B \ln\left(\frac{t}{t_B}\right) + \mathcal{O}(1) \quad (68)$$

Then the left-hand side behaves as $\ddot{u}_{\text{bg}} \sim q_B t^{-2}$, while the right-hand side is dominated by $\sim \Omega_B^2 C t^{-4q_B}$ with $C > 0$. Matching the leading powers gives

$$2 = 4q_B \quad \Rightarrow \quad q_B = \frac{1}{2} \quad (69)$$

This result is independent of any coefficient values; it follows from the equality of exponents alone. The asymptotic exponent of the radiation-dominated branch is therefore necessarily fixed to $q_B = 1/2$.

The same conclusion also follows from the first integral

$$(\dot{u}_{\text{bg}})^2 - \frac{\Omega_B^2}{2} e^{4(u_{\text{bg}} - u_B)} = C_0 \quad (70)$$

as well. Since the exponential term dominates as $t \rightarrow 0^+$, one obtains $u_{\text{bg}}(t) = -\frac{1}{2} \ln t + \mathcal{O}(1)$, confirming the same $q_B = 1/2$. This derivation fixes the baseline value of q_B used in the BBN freeze-out chain of Part III 5.5 (see Part IV for the verification materials).

Moreover, the first integral makes the exact family of solutions explicit. Defining $\Delta u \equiv u_{\text{bg}} - u_B$ and

$$z \equiv e^{-2\Delta u} \quad (71)$$

one obtains

$$\left(\frac{dz}{dt}\right)^2 = 2\Omega_B^2 + 4C_0 z^2 \quad (72)$$

The closed-form solutions then split according to the sign of C_0 :

For $C_0 > 0$,

$$z(\Delta t) = \frac{\Omega_B}{\sqrt{2C_0}} \sinh\left(2\sqrt{C_0} \Delta t\right) \quad (73)$$

For $C_0 = 0$,

$$z(\Delta t) = \sqrt{2} \Omega_B \Delta t \quad (74)$$

For $C_0 < 0$,

$$z(\Delta t) = \frac{\Omega_B}{\sqrt{-2C_0}} \sin\left(2\sqrt{-C_0} \Delta t\right) \quad (75)$$

where $\Delta t \equiv t - t_\star > 0$. In all cases, the expansion for $\Delta t \rightarrow 0^+$ is

$$z(\Delta t) = \sqrt{2} \Omega_B \Delta t + \mathcal{O}(\Delta t^3) \quad (76)$$

which is common, so

$$u_{\text{bg}}(t) = u_B - \frac{1}{2} \ln(\Delta t) + \mathcal{O}(\Delta t^2), \quad q_B = \frac{1}{2} \quad (77)$$

is fixed uniquely, independently of C_0 . Therefore $q_B = 1/2$ is not a mere asymptotic guess but a necessary consequence of the exact solution family of the background-wave equation with a blackbody-radiation source. The exact exponent $q_B = 1/2$ fixed in this radiation-dominated limit is used in Part III (quantum physics) to determine the weak-interaction freeze-out temperature in the early universe and to derive the minimal BBN chain without assuming spatial expansion, giving the He-4 mass fraction $Y_p \approx 0.25$. The final acceptance or rejection of the full multi-isotope consistency (D/H, He-3/He-4, Li-7) is governed by the Watch decision in Part III 5.5.2.

2.7.3 Full Time Evolution of the Background Wave $P_{\text{bg}}(t)$ (Radiation-Dominated -> Matter-Dominated -> Present Era)

The background field is not stopped at the radiation-dominated limit; it is defined so as to connect the matter-dominated branch and the present-era neighborhood under the same interface.

Define the background variables by

$$u_{\text{bg}}(t) \equiv \ln \left(\frac{P_{\text{bg}}(t)}{P_{\text{bg}}(t_0)} \right), \quad H_{\text{eff}}(t) \equiv -\dot{u}_{\text{bg}}(t) \quad (78)$$

and introduce the effective scale factor

$$a_{\text{eff}}(t) = \frac{P_{\text{bg}}(t_0)}{P_{\text{bg}}(t)} = e^{-u_{\text{bg}}(t)} \quad (79)$$

by

The continuity equation for each component,

$$\dot{\rho}_i + 3(1 + w_i)H_{\text{eff}}\rho_i = 0 \quad (80)$$

gives

$$\rho_i(u_{\text{bg}}) = \rho_{i,0} \exp(3(1 + w_i)u_{\text{bg}}) \quad (81)$$

where $w_r = 1/3$, $w_m = 0$, and $w_\Lambda = -1$.

Accordingly, the background equation

$$\ddot{u}_{\text{bg}} = 4\pi G \left(\rho + \frac{3p}{c^2} \right) \quad (82)$$

closes as

$$\ddot{u}_{\text{bg}} = \Omega_r^2 e^{4u_{\text{bg}}} + \Omega_m^2 e^{3u_{\text{bg}}} - \Omega_\Lambda^2 \quad (83)$$

with

$$\Omega_r^2 \equiv 8\pi G \rho_{r,0}, \quad \Omega_m^2 \equiv 4\pi G \rho_{m,0}, \quad \Omega_\Lambda^2 \equiv 8\pi G \rho_{\Lambda,0}. \quad (84)$$

The first integral is

$$\dot{u}_{\text{bg}}^2 = H_0^2 + \frac{\Omega_r^2}{2}(e^{4u_{\text{bg}}} - 1) + \frac{2\Omega_m^2}{3}(e^{3u_{\text{bg}}} - 1) - 2\Omega_\Lambda^2 u_{\text{bg}} \quad (85)$$

and therefore the complete solution is

$$t - t_0 = \int_0^{u_{\text{bg}}(t)} \frac{d\tilde{u}}{\sqrt{H_0^2 + \frac{\Omega_r^2}{2}(e^{4\tilde{u}} - 1) + \frac{2\Omega_m^2}{3}(e^{3\tilde{u}} - 1) - 2\Omega_\Lambda^2 \tilde{u}}} \quad (86)$$

which is defined uniquely by quadrature.

The direct form for observables, using $1 + z = e^{u_{\text{bg}}}$, is

$$H_{\text{eff}}^2(z) = H_0^2 + \frac{\Omega_r^2}{2}[(1+z)^4 - 1] + \frac{2\Omega_m^2}{3}[(1+z)^3 - 1] - 2\Omega_\Lambda^2 \ln(1+z) \quad (87)$$

as follows.

To fix the high-redshift behavior, determine analytically how strongly the $\ln(1+z)$ term contributes at high redshift. Define the separate contributions by

$$X_r(z) \equiv \frac{\Omega_r^2}{2}[(1+z)^4 - 1], \quad X_m(z) \equiv \frac{2\Omega_m^2}{3}[(1+z)^3 - 1], \quad X_\Lambda(z) \equiv -2\Omega_\Lambda^2 \ln(1+z) \quad (88)$$

Then for $z \gg 1$,

$$\frac{|X_\Lambda(z)|}{X_m(z)} \sim \frac{3\Omega_\Lambda^2}{\Omega_m^2} \frac{\ln(1+z)}{(1+z)^3} \rightarrow 0, \quad \frac{|X_\Lambda(z)|}{X_r(z)} \sim \frac{4\Omega_\Lambda^2}{\Omega_r^2} \frac{\ln(1+z)}{(1+z)^4} \rightarrow 0 \quad (89)$$

Thus the $\ln(1+z)$ term does not become dominant at high z ; the asymptotic behavior is controlled by the radiation term $\propto (1+z)^4$ and the matter term $\propto (1+z)^3$. This is also consistent with the tendency, in the Pantheon+/CC range ($z \lesssim 2$), for the $\ln(1+z)$ term to be partially degenerate with the matter term, as seen in the full-form refit of Part II 5.3 where $\alpha_\Lambda \rightarrow 0$.

For a single component with $w > -1$, the closed form

$$P_{\text{bg}}^{(w)}(t) = P_* \left[1 + \frac{3}{2}(1+w)H_*(t-t_*) \right]^{-\frac{2}{3(1+w)}} \quad (90)$$

has the exponent

$$q(w) = \frac{2}{3(1+w)} \quad (91)$$

In particular,

$$q_r = \frac{1}{2}, \quad q_m = \frac{2}{3} \quad (92)$$

which is consistent with the radiation-dominated derivation of Section 2.6.2, namely $q_B = 1/2$.

Near the present epoch, where the Λ term dominates, use

$$u_{\text{bg}}(t) \approx -H_\Lambda(t-t_\Lambda) - \frac{\Omega_\Lambda^2}{2}(t-t_\Lambda)^2, \quad (93)$$

$$P_{\text{bg}}(t) \approx P_{\Lambda} \exp \left[-H_{\Lambda}(t - t_{\Lambda}) - \frac{\Omega_{\Lambda}^2}{2}(t - t_{\Lambda})^2 \right] \quad (94)$$

where P_{Λ} and H_{Λ} are fixed at the junction point t_{Λ} to the matter branch by the C^1 continuity conditions, namely continuity of P and $\dot{P}$.

In this way, the background field $P_{\text{bg}}(t)$ is defined as a complete time evolution that connects the radiation branch ($q = 1/2$), the matter branch ($q = 2/3$), and the present-era neighborhood (the Λ term) within the same system of equations.

2.8 Reference Action Skeleton (Reference Definition), Rotation (Frame Dragging), and the Four-Vector Extension

In this section, the action skeleton and Noether conservation law are first fixed as Part I reference definitions. On that basis, the minimal covariant extension is defined so as to describe rotational mass effects (frame dragging) while preserving the static scalar branch.

2.8.1 Reference Action Skeleton and Noether Conservation Law (Part I Reference Definition)

In this paper, the **reference action skeleton is fixed in Part I**. By "reference skeleton" we mean the skeleton that explicitly defines the P-model's own derived terms (the scalar kinetic term $\mathcal{L}_\chi$, the free vector propagation term $\mathcal{L}_{P_\mu}^{\text{free}}$, and the minimal coupling term $\mathcal{L}_{\text{int}}$), while for the matter, electromagnetic, and rotational sectors it declares only the structural roles as adopted effective sectors. The derivation status and acceptance or rejection of those adopted sectors are governed by Parts III-A/B.

Reference action skeleton (minimal form):

$$S_{\text{total}} := \int d^4x \mathcal{L}_{\text{total}}^{\text{vec}} \quad (95)$$

$$\begin{aligned} \mathcal{L}_{\text{total}}^{\text{vec}} &:= \mathcal{L}_\chi + \mathcal{L}_{P_\mu}^{\text{free}} + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{rot}}, \\ \chi &\equiv \ln(P_t/P_{\text{ref}}), \quad \phi \equiv -c^2 \chi \end{aligned} \quad (96)$$

$$\mathcal{L}_\chi := \frac{M_\chi^2}{2} \partial_\mu \chi \partial^\mu \chi - U(\chi), \quad F_{\mu\nu}^{(P)} \equiv \partial_\mu P_\nu - \partial_\nu P_\mu \quad (97)$$

$$\mathcal{L}_{P_\mu}^{\text{free}} := -\frac{1}{4} F_{\mu\nu}^{(P)} F^{\mu\nu}_{(P)} + \frac{m_P^2}{2} P_\mu P^\mu \quad (98)$$

The Proca-type expression written here is the minimal form used to display the action skeleton in the shortest possible way. The complete form including gauge closure is fixed in Section 2.8.2, and the above expression is recovered in the limit $Z_P = 1$, Stückelberg field $\pi = 0$, and $\xi_g \rightarrow \infty$.

The minimal coupling to matter is defined by

$$\mathcal{L}_{\text{int}} = g_P P_\mu J_{\text{matter}}^\mu \quad (99)$$

and in the particle limit we adopt

$$S_a := -m_a c \int e^{-\chi} \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} \quad (100)$$

which directly recovers

$$\frac{d\tau}{dt} = \frac{P_{\text{ref}}}{P_t} \sqrt{1 - v^2/c^2} \quad (101)$$

In the optical limit, take

$$n(P_t) = \left(\frac{P_t}{P_{\text{ref}}} \right)^{2\beta} \quad (102)$$

which connects the theory to the mappings for Shapiro delay and light bending.

Status of each term:

- $\mathcal{L}_\chi$, $\mathcal{L}_{P_\mu}^{\text{free}}$, and $\mathcal{L}_{\text{int}}$: P-model-specific derived terms explicitly defined in this section. Here $\chi \equiv \ln(P_t/P_{\text{ref}})$ is a derived quantity rewritten from P_t and is not an independent fundamental field.
- $\mathcal{L}_{\text{matter}}$: the standard matter-field action is adopted. In this paper the particle-limit action S_a is kept as the minimal form, while the full quantum-field-theoretic form is delegated to Part III-A.
- $\mathcal{L}_{\text{EM}}$: the photon branch is derived from the transverse component of P_μ (Part III-A 2.6). This paper declares only the structural connection; the derivation and its acceptance or rejection are governed by Part III-A.
- $\mathcal{L}_{\text{rot}}$: at the present stage the full Lagrangian density has not been fixed uniquely. Only the structure needed for the frame-dragging mapping is fixed in Section 2.8.7, while candidate microscopic extensions are isolated in the reference note of Section 2.8.2 and judged in Part III-B. This does not imply a failure of the action skeleton.

Noether conservation law (fixed form):

$$T_{\text{tot}}^{\mu\nu} := \sum_i \frac{\partial \mathcal{L}_{\text{total}}^{\text{vec}}}{\partial(\partial_\mu \Phi_i)} \partial^\nu \Phi_i - \eta^{\mu\nu} \mathcal{L}_{\text{total}}^{\text{vec}}, \quad \partial_\mu T_{\text{tot}}^{\mu\nu} = 0 \quad (103)$$

Here Φ_i denotes P_μ and matter. $\chi \equiv \ln(P_t/P_{\text{ref}})$ is a derived quantity fixed by P_t and is not an independent fundamental degree of freedom. The optical degree of freedom $A_\mu^{(P)}$ is derived as the transverse component of P_μ and is likewise not an independent fundamental degree of freedom. In addition, for the local phase U(1) retained as an adopted effective sector, fix

$$\partial_\mu J^\mu = 0 \quad (104)$$

as the reference conservation law.

Uniqueness conditions (frozen):

- Fundamental degrees of freedom: $\{P_\mu, \text{matter}\}$ (here $\chi \equiv \ln(P_t/P_{\text{ref}})$ is derived from P_t , and the optical degree of freedom $A_\mu^{(P)} \equiv \delta P_\mu^T / \sqrt{Z_P}$ is likewise derived as the transverse component of P_μ)
- Differential order: up to second order only (higher derivatives are not adopted)
- Symmetries: translation and rotation (intrinsic to the P-model); local phase U(1) is retained as an adopted effective sector, while the derivation of its origin is delegated to Part III-A 3.2

With these conditions, the main equations of Part I (gravitational acceleration, clock rate, and the light-propagation mapping) can be tracked self-consistently as mappings from the reference action skeleton.

The same skeleton also fixes the localization condition for the bound states used in the weak-interaction sector. When the photon is separated out as the transverse component $A_\mu^{(P)} = \delta P_\mu^T / \sqrt{Z_P}$, the

remaining non-transverse components $\{\delta P_0, \delta P_i^L\}$ decompose into one propagating massive mode and one constrained mode. Therefore, the existence of a bound state is judged not by the separate-component condition $\kappa_n^2 = 1 - \beta_n^2$ but by the damped eigenmode of the coupled system.

$$\kappa_{\text{coupled}}^2 = m_0^2 - \beta_n^2, \quad m_0^2 = \frac{4\lambda v^2}{Z_P} \quad (105)$$

Even when the separate components appear non-localized because $\beta_n > 1$, they are treated as a physically allowed two-component bound state as long as the coupled system satisfies $\kappa_{\text{coupled}} > 0$. The apparent non-localization of the individual components is a spurious effect of ignoring the coupling; the final localization decision is based on the coupled eigenmode.

The same frozen-action skeleton also provides the derivational anchor that selects the P-model's own fine-structure constant without free parameters. Specifically, after separating out the photon branch from the transverse component of P_μ , one can construct two independent readouts from the remaining vector Q-ball family, and the consistency condition between those readouts selects the P-model's fine-structure constant (Part III-A 2.6). Part I limits itself to defining the frozen action skeleton that serves as the starting point of that derivation; the derivation itself, its numerical audit, and the residual evaluation are governed by Part III-A. Likewise, the theoretical anchor point for the W/Z boson masses in the two-component coupled Q-ball picture is obtained from the same frozen-action skeleton (Part III-A 2.8 / Part III-B 5.5). The canonical derivations and the acceptance or rejection of these items are governed by Parts III-A/B.

2.8.2 Minimal Coupling Between the Four-Vector P_μ and the Matter Current J^μ

Define the time wave by

$$P_\mu = (P_t, P_1, P_2, P_3), \quad J_{\text{matter}}^\mu = (\rho c, \rho \mathbf{v}) \quad (106)$$

To satisfy both local gauge invariance as a vector field and boundedness of the Hamiltonian (elimination of ghost modes), define the minimal closed action by introducing a Stückelberg field π . Here π is an auxiliary field that restores gauge invariance and is not a physical fundamental degree of freedom; it is removed in the unitary gauge $\pi = 0$.

$$F_{\mu\nu}^{(P)} \equiv \partial_\mu P_\nu - \partial_\nu P_\mu \quad (107)$$

$$\mathcal{L}_{P_\mu}^{\text{free}} = -\frac{Z_P}{4} F_{\mu\nu}^{(P)} F^{\mu\nu}_{(P)} + \frac{m_P^2}{2} \left(P_\mu - \frac{1}{m_P} \partial_\mu \pi \right) \left(P^\mu - \frac{1}{m_P} \partial^\mu \pi \right) - \frac{1}{2\xi_g} (\partial_\mu P^\mu + \xi_g m_P \pi)^2 \quad (108)$$

and take the minimal interaction term with matter to be

$$\mathcal{L}_{\text{int}} = g_P P_\mu J_{\text{matter}}^\mu \quad (109)$$

In this paper, the Stückelberg-plus-gauge-fixing form is adopted as the frozen canonical form that includes gauge closure. Within the reference action skeleton, the minimal form involving the vector coupling is

$$\mathcal{L}_{\text{total}}^{\text{vec}} = \mathcal{L}_\chi + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{EM}} + \mathcal{L}_{P_\mu}^{\text{free}} + \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{rot}} \quad (110)$$

while, within the reference skeleton fixed in Section 2.8.1, the rotational coupling is carried by $\mathcal{L}_{P_\mu}^{\text{free}}, \mathcal{L}_{\text{int}}, \mathcal{L}_{\text{rot}}$. However, the Lagrangian density of $\mathcal{L}_{\text{rot}}$ itself has not yet been fixed uniquely at the present stage. Here it is retained only as an effective-sector label representing the structure needed for the frame-dragging mapping. This does not mean that the action skeleton has failed; the rotational sector is kept as an effective-sector label, and the structure needed for the frame-dragging mapping is already fixed in Section 2.8.7. Accordingly, the Noether conservation law of the energy-momentum tensor,

$$\partial_\mu T_{\text{tot}}^{\mu\nu} = 0 \quad (111)$$

is adopted as the reference form under the frozen action skeleton. Stable dynamics, whether macroscopic or microscopic, are traced under this Noether form (for the quantum-side connection see Part III 2.7).

Reference note (candidate microscopic dictionary): As a candidate extension of the macroscopic rotation coupling λ_{rot} to a microscopic current, consider the coupling to the left-handed chiral fermion current J_L^μ ,

$$\mathcal{L}_{\text{int}} + \mathcal{L}_{\text{rot}} \supset -\lambda_{\text{rot}} g_P \bar{\psi} \gamma^\mu \frac{1 - \gamma^5}{2} \psi P_\mu - \frac{\lambda_{\text{rot}} g_P}{4m} \bar{\psi} \sigma^{\mu\nu} \psi F_{\mu\nu}^{(P)} \quad (112)$$

This candidate yields frame dragging for rotating stars on the macroscopic side and provides a minimal candidate reconstruction of the V-A structure of the weak interaction and the tensor term of the nuclear force on the microscopic side. However, it is not a coupling automatically derived from P; it is a hand-imposed minimal candidate dictionary. At the present stage, numerical audit rejects Route B (the P-model's own V-A mechanism), and its formal acceptance or rejection is governed by Part III-B 5.5.

2.8.3 1A Algebraic Bridge from the Vector Field to the Effective Metric $g_{\mu\nu}(P)$ (caseB)

P_μ is mapped to the effective metric $g_{\mu\nu}(P)$ through the following four steps.

First, decompose the four-vector as

$$P_\mu = P_t U_\mu + P_\mu^\perp, \quad U_\mu U^\mu = -1, \quad U^\mu P_\mu^\perp = 0 \quad (113)$$

orthogonally, and

$$u \equiv \ln\left(\frac{P_t}{P_{\text{ref}}}\right), \quad \phi \equiv -c^2 u \quad (114)$$

define

Next, define the symmetric rank-2 tensor (the effective carrier of the spin-2 observable) by

$$h_{\mu\nu} \equiv \eta_{\mu\nu} + U_\mu U_\nu \quad (115)$$

$$g_{\mu\nu}^{(P)} := e^{-2u} U_\mu U_\nu + e^{2u} h_{\mu\nu} - \xi_{\text{rot}} \left(U_\mu P_\nu^\perp + U_\nu P_\mu^\perp \right) + \mathcal{O}\left((P^\perp)^2\right) \quad (116)$$

and fix

$$h_{\mu\nu}^{(2)}(P) \equiv g_{\mu\nu}^{(P)} - \eta_{\mu\nu} \quad (117)$$

as the effective spin-2 component generated algebraically from P_μ . In the comoving frame ($U^\mu = (1, 0, 0, 0)$), one has

$$g_{tt}^{(P)} = -e^{-2u}, \quad g_{ti}^{(P)} = -\xi_{\text{rot}} P_i, \quad g_{ij}^{(P)} = e^{2u} \delta_{ij} + \mathcal{O}\left((P^\perp)^2\right) \quad (118)$$

which gives

The mapping to observables is

$$a_i = -\frac{c^2}{2} \partial_i \ln(-g_{tt}^{(P)}) = c^2 \partial_i u = -\partial_i \phi \quad (119)$$

$$\beta_i^{(P)} \equiv -\frac{g_{ti}^{(P)}}{g_{tt}^{(P)}}, \quad \Omega_{\text{drag}}^{(P)} = \frac{1}{2} \left| \nabla \times \boldsymbol{\beta}^{(P)} \right| \quad (120)$$

which closes the algebraic bridge. On the optical side,

$$g_{\mu\nu}^{\text{opt}} = g_{\mu\nu}^{(P)} + (1 - n(P_t)^{-2}) U_\mu U_\nu, \quad n(P_t) = \left(\frac{P_t}{P_\infty} \right)^{2\beta} \quad (121)$$

connects the construction uniquely to the refraction mapping of Section 2.4.

2.8.4 1A.1 Deriving $\beta = 1$ from the Null Geodesics of the Effective Metric (Theoretical Prediction)

As the theoretical fixing, identify the far-field reference with $P_{\text{ref}} \equiv P_\infty$ and adopt the isotropic effective metric (caseB)

$$ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} (dr^2 + r^2 d\Omega^2), \quad u = \ln\left(\frac{P_t}{P_\infty}\right) \quad (122)$$

taking its null geodesics ($ds^2 = 0$). This gives

$$c_{\text{eff}} \equiv \frac{dr}{dt} = c e^{-2u} = c \left(\frac{P_t}{P_\infty} \right)^{-2} \quad (123)$$

Hence the refractive index is

$$n \equiv \frac{c}{c_{\text{eff}}} = e^{2u} = \left(\frac{P_t}{P_\infty} \right)^2 \quad (124)$$

which, compared with the general optical form

$$n(P_t) = \left(\frac{P_t}{P_\infty} \right)^{2\beta} \quad (125)$$

yields

$$\beta = 1 \quad (126)$$

Thus, in this paper, β is not an arbitrary adjustment parameter. It is fixed as a theoretical prediction derived from the isotropic effective-metric condition $g_{tt} \cdot g_{rr} = -c^2$ (or -1 in units with $c = 1$).

The conditions for the existence of a difference prediction relative to GR are defined by

$$\Delta_{\mu\nu} \equiv g_{\mu\nu}^{(P)} - g_{\mu\nu}^{(\text{GR})} \quad (127)$$

and the decision rule of this paper is fixed as follows.

$$C_{\text{weak}} : \Delta_{\mu\nu}^{(1\text{PN})} \approx 0 \iff |z_\gamma| \leq 1, |\mu - 1| \leq 1\sigma \quad (128)$$

$$C_{\text{diff}} : \Delta_{ij}^{(2)} \neq 0 \iff \mathcal{N}_0^{(2)} \neq 0, \lambda_H^{(N0)} > 0 \quad (129)$$

That is, caseB is fixed as the branch that agrees with the reference framework at weak-field 1PN order (C_{weak}) while still carrying a difference prediction in the second-order nonlinear terms (C_{diff}).

2.8.5 1B Gauge Closure and Ghost Elimination for P_μ (Moved from Part III 2.7.1)

To consolidate the closure definition of the four-vector field P_μ into the Part I baseline, fix the minimal closed action as

$$\begin{aligned} \mathcal{L}_{P,\text{full}} := & -\frac{Z_P}{4} F_{\mu\nu}^{(P)} F_{(P)}^{\mu\nu} + \frac{m_P^2}{2} \left(P_\mu - \frac{1}{m_P} \partial_\mu \pi \right) \left(P^\mu - \frac{1}{m_P} \partial^\mu \pi \right) \\ & - \frac{1}{2\xi_g} (\partial_\mu P^\mu + \xi_g m_P \pi)^2 + g_P P_\mu J_{\text{matter}}^\mu + \lambda_{\text{rot}} \mathcal{O}_{\text{spin}}[P_\mu, J_{\text{matter}}^\mu] \end{aligned} \quad (130)$$

with gauge transformation

$$P_\mu \rightarrow P_\mu + \partial_\mu \alpha, \quad \pi \rightarrow \pi + m_P \alpha \quad (131)$$

This defines the gauge sector.

Writing the propagator obtained from the quadratic action as

$$D_{\mu\nu}(k) = \frac{-i}{Z_P} \left[\frac{\eta_{\mu\nu} - k_\mu k_\nu / m_P^2}{k^2 - m_P^2 + i0} + \frac{\xi_g k_\mu k_\nu / m_P^2}{k^2 - \xi_g m_P^2 + i0} \right] \quad (132)$$

then throughout the parameter region $Z_P > 0$, $m_P^2 \geq 0$, $\xi_g > 0$, the residues are non-negative and no negative-norm (ghost) mode appears.

The Noether energy-momentum tensor is

$$T_{(P)}^{\mu\nu} := -Z_P F^{(P)\mu\alpha} F_{(P)\alpha}^\nu + \frac{Z_P}{4} \eta^{\mu\nu} F_{\alpha\beta}^{(P)} F_{(P)}^{\alpha\beta} + m_P^2 \Pi^\mu \Pi^\nu - \eta^{\mu\nu} \frac{m_P^2}{2} \Pi_\alpha \Pi^\alpha + \dots \quad (133)$$

where $\Pi_\mu := P_\mu - \partial_\mu \pi / m_P$. Using the field equations, one obtains

$$\partial_\mu T_{\text{tot}}^{\mu\nu} = 0 \quad (134)$$

as the closure relation.

2.8.6 Exact Reduction in the Stationary Limit (Agreement with the Existing Scalar Theory)

For stationary matter, require $J^i = 0$. The spatial components then reduce to

$$J^i = 0 \Rightarrow P_i = 0, \quad P_t \equiv P \quad (135)$$

which recovers exactly the scalar mappings fixed in Sections 2.1-2.4,

$$\phi = -c^2 \ln\left(\frac{P}{P_\infty}\right), \quad a = c^2 \nabla \ln\left(\frac{P}{P_\infty}\right), \quad \frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt}\right)_v, \quad n(P) = \left(\frac{P}{P_\infty}\right)^{2\beta} \quad (136)$$

Therefore the already-established pass region is preserved non-destructively in the stationary limit.

2.8.7 Frame-Dragging Mapping (Connection to Observation)

For a rotating source, $J^i \neq 0$ and the component P_i becomes nonzero. First fix the quasi-static equation for the $\nu = \phi$ component as

$$\frac{1}{r^2} \partial_r (r^2 \partial_r P_\phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta P_\phi) - \frac{P_\phi}{r^2 \sin^2 \theta} = -\frac{g_P}{c^2} J_\phi \quad (137)$$

The exterior solution, valid when r is sufficiently larger than the source radius, is

$$P_\phi(r, \theta) = \frac{g_P J_*}{2c^2} \frac{\sin^2 \theta}{r^2} + \mathcal{O}(r^{-3}) \equiv \frac{\mu_{\text{drag}} \sin^2 \theta}{r^2}, \quad \mu_{\text{drag}} \equiv \frac{g_P J_*}{2c^2} \quad (138)$$

where J_* is the angular momentum of the source. Next define the off-diagonal component of the effective metric from P_ϕ as

$$ds^2 = g_{tt}^{(P)} dt^2 + 2g_{t\phi}^{(P)} dt d\phi + \dots, \quad g_{t\phi}^{(P)} \equiv -g_{tt}^{(P)} \beta_\phi^{(P)}, \quad \beta_\phi^{(P)} \equiv \xi_{\text{rot}} P_\phi \quad (139)$$

This closes the mapping $P_\phi \rightarrow g_{t\phi}^{(P)}$ uniquely. Now take the parallel transport of the spin four-vector S^μ ,

$$U^\mu \nabla_\mu S^\nu = 0, \quad U_\mu S^\mu = 0 \quad (140)$$

and reduce it to the weak-field, low-velocity limit:

$$\frac{d\mathbf{S}}{dt} = \boldsymbol{\Omega}_{\text{drag}}^{(P)} \times \mathbf{S}, \quad \boldsymbol{\Omega}_{\text{drag}}^{(P)} \equiv \frac{1}{2} \nabla \times \boldsymbol{\beta}^{(P)} \quad (141)$$

Substituting $\beta_\phi^{(P)} = \xi_{\text{rot}} P_\phi$ gives

$$(\nabla \times \boldsymbol{\beta}^{(P)})_r = \frac{1}{r \sin \theta} \partial_\theta (\sin \theta \beta_\phi^{(P)}) = \frac{3\xi_{\text{rot}} \mu_{\text{drag}} \sin \theta \cos \theta}{r^3}, \quad (142)$$

$$(\nabla \times \boldsymbol{\beta}^{(P)})_\theta = -\frac{1}{r} \partial_r (r \beta_\phi^{(P)}) = \frac{\xi_{\text{rot}} \mu_{\text{drag}} \sin^2 \theta}{r^3} \quad (143)$$

and on the equatorial plane ($\theta = \pi/2$),

$$\Omega_{\text{pred}}^{(P)} = \left| \Omega_{\text{drag}}^{(P)} \right| = \frac{|\xi_{\text{rot}} \mu_{\text{drag}}|}{2r^3} = \frac{|g_P \xi_{\text{rot}} J_*|}{4c^2 r^3} \quad (144)$$

Thus the chain $P_\phi \rightarrow g_{t\phi}^{(P)} \rightarrow \Omega_{\text{pred}}^{(P)}$ alone yields the r^{-3} precession law. The observational comparison is performed through

$$\mathcal{R}_{\text{drag}} \equiv \frac{|\Omega_{\text{obs}}|}{|\Omega_{\text{pred}}^{(P)}|} \quad (145)$$

If $\mathcal{R}_{\text{drag}} \approx 1$ is obtained, the theory is consistent with weak-field rotation measurements. The numerical audit against the primary GP-B / LAGEOS data is collected in Part II 4.12, while derivational closure and rejection conditions are collected in Part IV Section 3.[8][9]

Reference note: the derivation in this section is closed from P_μ and the effective metric $g_{\mu\nu}(P)$ alone. The reference framework is used only in the final comparison step, namely in deciding agreement with the observed values.

2.8.8 Minimal Derivation of the Rotating Strong-Field Vacuum Solution (Stationary Axisymmetry)

Fix the Euler-Lagrange equation derived from $\mathcal{L}_{P_\mu}^{\text{free}}$ in the vacuum limit ($J^\mu = 0$) and the massless limit $m_P \rightarrow 0$ as

$$\partial_\mu F_{(P)}^{\mu\nu} = 0, \quad F_{\mu\nu}^{(P)} = \partial_\mu P_\nu - \partial_\nu P_\mu \quad (146)$$

The result is the above equation. The choice of background metric is compared on the same I/F through caseA/caseB:

$$\text{caseA} : \eta_{\mu\nu} \Rightarrow \text{reject}, \quad \text{caseB} : g_{\mu\nu}(P) \Rightarrow \text{pass} \quad (147)$$

The metric selection is fixed by adopting the effective-metric branch. Therefore the regular vacuum equation used in this paper is

$$\nabla_\mu^{(g(P))} F_{(P)}^{\mu\nu} = 0, \quad \frac{1}{\sqrt{|g(P)|}} \partial_\mu \left(\sqrt{|g(P)|} F_{(P)}^{\mu\nu} \right) = 0 \quad (148)$$

as follows. The minimal stationary axisymmetric branch is

$$ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} (dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2), \quad u \equiv \ln \left(\frac{P_t}{P_{\text{ref}}} \right) \quad (149)$$

$$\sqrt{|g(P)|} = e^{2u} r^2 \sin \theta, \quad F_{(P)}^{\mu\nu} = g^{\mu\alpha}(P) g^{\nu\beta}(P) (\partial_\alpha P_\beta - \partial_\beta P_\alpha) \quad (150)$$

with

$$\partial_t = 0, \quad \partial_\phi = 0, \quad P_\mu = (P_t(r, \theta), 0, 0, P_\phi(r, \theta)) \quad (151)$$

Assuming this form, the independent components reduce to the following nonlinear coupled partial

differential equations.

$$\frac{1}{r^2} \partial_r (r^2 e^{-2u} \partial_r P_t) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta e^{-2u} \partial_\theta P_t) = \mathcal{N}_0^{(2)}[u, P_t, P_\phi] \quad (152)$$

$$\frac{1}{r^2} \partial_r (r^2 e^{-2u} \partial_r P_\phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta e^{-2u} \partial_\theta P_\phi) - \frac{e^{-2u} P_\phi}{r^2 \sin^2 \theta} = \mathcal{N}_\phi^{(2)}[u, P_t, P_\phi] \quad (153)$$

Here $\mathcal{N}_0^{(2)}$ denotes the quadratic source term in the P_t equation, while $\mathcal{N}_\phi^{(2)}$ denotes the quadratic source term in the rotational component P_ϕ equation. In addition, the rotation-induced contribution inside the P_t equation is written as $\mathcal{N}_{0,\phi}^{(2)}$ in order to distinguish it from $\mathcal{N}_\phi^{(2)}$ on the P_ϕ side.

Let $u = \epsilon\psi$ with $\epsilon \ll 1$. The explicit $\mathcal{O}(\epsilon^2)$ terms are fixed as

$$\mathcal{N}_0^{(2)} = \mathcal{N}_{\text{metric}}^{(2)} + \mathcal{N}_{0,\phi}^{(2)} \quad (154)$$

$$\mathcal{N}_{\text{metric}}^{(2)} = \epsilon_{\text{nl}}^2 2 \left[(\partial_r u)(\partial_r P_t) + \frac{1}{r^2} (\partial_\theta u)(\partial_\theta P_t) \right] \quad (155)$$

$$\mathcal{N}_{0,\phi}^{(2)} = \epsilon_{\text{nl}}^2 e^{-2u} \left[\frac{(\partial_r P_\phi)^2}{r^2 \sin^2 \theta} + \frac{(\partial_\theta P_\phi)^2}{r^4 \sin^2 \theta} \right] \quad (156)$$

with the minimal-branch rotational term $P_\phi(r, \theta) = \mu_{\text{drag}} \sin^2 \theta / r^2$.

Furthermore, adopt

$$P_t = P_t^{(\text{lin})} + \delta P_t, \quad \bar{N}_0(r) = \frac{1}{2} \int_0^\pi \mathcal{N}_0^{(2)}(r, \theta) \sin \theta d\theta \quad (157)$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d \delta P_t}{dr} \right) = \bar{N}_0(r), \quad \delta P_t(r) = - \int_r^{r_{\text{out}}} d\xi \xi^{-2} \int_\xi^{r_{\text{out}}} ds s^2 \bar{N}_0(s) \quad (158)$$

as the perturbative solution, with $\delta P_t(r_{\text{out}}) = 0$.

The strong-field higher-order coefficient is interpreted not as a fit coefficient but as something fixed by the strength of $\mathcal{N}_0^{(2)}$.

$$\Delta C_{\text{core}} \equiv C_{\text{linear}} - C_{\text{ref}}, \quad \Delta C_{N0} \equiv C_{\text{full}} - C_{\text{linear}} \quad (159)$$

$$\lambda_H^{(N0)} \equiv \frac{\Delta C_{N0}}{\Delta C_{\text{core}}} = \frac{0.1806\%}{4.6278\%} \quad (160)$$

$$\lambda_H^{(N0)} = \epsilon_{\text{nl}}^2 \Xi_{N0} \quad (161)$$

Under this definition, λ_H is uniquely fixed as the dimensionless strength measuring how much $N_0^{(2)}$ contributes to the strong-field coefficient difference. From the fixed values $\Delta C_{\text{core}} = 4.6278\%$, $\Delta C_{N0} = 0.1806\%$, and $\epsilon_{\text{nl}}^2 = 0.0064$, one obtains $\lambda_H^{(N0)} = 0.0390$ and $\Xi_{N0} = 6.10$, which connect directly to the Part II / Part IV audit I/F.

The approximation order is

$$e^{-2u} = 1 - 2u + 2u^2 + \mathcal{O}(u^3) \quad (162)$$

That is, terms are retained through $\mathcal{O}(\epsilon^2)$ and those of order $\mathcal{O}(\epsilon^3)$ and higher are discarded. The boundary conditions are

$$r \rightarrow \infty : \frac{P_t}{P_{\text{ref}}} \rightarrow 1 + \frac{1}{r} + \frac{\lambda_2}{r^2}, \quad P_\phi \rightarrow 0 \quad (163)$$

$$\theta = 0, \pi : P_\phi = 0, \quad \partial_\theta P_t = 0, \quad r \rightarrow r_H^+ : P_t, P_\phi \text{ finite (finite radial flux)} \quad (164)$$

$$\theta = \frac{\pi}{2} : \partial_\theta P_t = 0 \quad (165)$$

These are adopted, and the closure decision is fixed as

$$C_{\text{closure}} = C_{\text{axis}} \wedge C_{\text{asympt}} \wedge C_{\text{horizon}} \wedge (\sigma_{\text{flux,rel}} \leq 1.0 \times 10^{-3}) \quad (166)$$

using the observed fixed value $\sigma_{\text{flux,rel}} = 1.2527 \times 10^{-16}$. The correspondence with machine-readable decision keys is collected in the Part IV audit registry. The following linear forms are retained as auxiliary mappings in the weak-field far-zone limit where $g(P) \rightarrow \eta$ and $\mathcal{N}_\nu^{(2)} \rightarrow 0$.

$$\frac{1}{r^2} \partial_r (r^2 \partial_r P_t) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta P_t) = 0 \quad (167)$$

$$\frac{1}{r^2} \partial_r (r^2 \partial_r P_\phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta P_\phi) - \frac{P_\phi}{r^2 \sin^2 \theta} = 0 \quad (168)$$

Using the separable form

$$P_t = R_\ell^{(0)}(r) \Theta_\ell^{(0)}(\theta), \quad P_\phi = R_\ell^{(\phi)}(r) \Theta_\ell^{(\phi)}(\theta) \quad (169)$$

one obtains the common radial equation

$$\frac{d}{dr} \left(r^2 \frac{dR_\ell}{dr} \right) - \ell(\ell+1) R_\ell = 0 \quad (170)$$

and the angular equations

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta_\ell^{(0)}}{d\theta} \right) + \ell(\ell+1) \Theta_\ell^{(0)} = 0 \quad (171)$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta_\ell^{(\phi)}}{d\theta} \right) - \frac{\Theta_\ell^{(\phi)}}{\sin^2 \theta} + \ell(\ell+1) \Theta_\ell^{(\phi)} = 0 \quad (172)$$

These imply that the general external solution has the form

$$P_t(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + B_\ell r^{-\ell-1} \right) P_\ell(\cos \theta) \quad (173)$$

$$P_\phi(r, \theta) = \sum_{\ell=1}^{\infty} \left(C_\ell r^\ell + D_\ell r^{-\ell-1} \right) P_\ell^1(\cos \theta) \quad (174)$$

as shown above. Operationally, the minimal external branch is fixed by $P_t/P_{\text{ref}} = 1 + a_1/r + a_2/r^2 + \dots$ and $P_\phi = (\mu_{\text{drag}} \sin^2 \theta)/r^2 + \dots$, for which the rotational term derived from P_ϕ gives $\Omega_{\text{drag}} \propto r^{-3}$. This branch is adopted as the foundation for the strong-field audit (EHT/GW), while its acceptance or rejection is decided on the same I/F in Part II. The fixed values in the weak-field coefficient audit are Mercury coefficient 0.3333298333 in caseA (reject) and Mercury coefficient 1.0000105000 with $z_\gamma = -0.7346$ and nonlinear PDE closure in caseB (pass).

2.8.9 First-Principles Prediction of the EHT Mapping Coefficient κ (Radiative Transport Under the Effective Metric)

The EHT coefficient κ (ring diameter / shadow diameter) is not fixed by the capture boundary $b_{\text{sh}}(P)$ (geometry) alone. It is fixed by simultaneously solving light-ray tracing on $g_{\mu\nu}(P)$ and the radiative transfer of emission/absorption using P_μ and J^μ .

Fix the effective optical metric by

$$g_{\mu\nu}^{\text{opt}} = g_{\mu\nu}(P) + (1 - n(P)^{-2}) U_\mu U_\nu, \quad n(P) = \left(\frac{P}{P_\infty} \right)^{2\beta} \quad (175)$$

Then the rays satisfy

$$g_{\text{opt}}^{\mu\nu} k_\mu k_\nu = 0, \quad \frac{dk_\mu}{d\lambda} = -\frac{1}{2} \partial_\mu g_{\text{opt}}^{\alpha\beta} k_\alpha k_\beta \quad (176)$$

as above. The radiative transfer is written in terms of the invariants

$$\tilde{I}_\nu \equiv \frac{I_\nu}{\nu^3}, \quad \tilde{J}_\nu \equiv \frac{j_\nu}{\nu^2}, \quad \tilde{A}_\nu \equiv \alpha_\nu \nu \quad (177)$$

and

$$k^\mu \nabla_\mu^{(g(P))} \tilde{I}_\nu = \tilde{J}_\nu - \tilde{A}_\nu \tilde{I}_\nu \quad (178)$$

Extending this to polarization, the Stokes vector

$$\tilde{\mathbf{S}}_\nu \equiv (\tilde{I}_\nu, \tilde{Q}_\nu, \tilde{U}_\nu)^T \quad (179)$$

obeys

$$k^\mu \nabla_\mu^{(g(P))} \tilde{\mathbf{S}}_\nu = \tilde{\mathbf{J}}_\nu - \tilde{\mathbf{A}}_\nu \tilde{\mathbf{S}}_\nu + \dot{\tau}_T \mathbf{M}(\mu, \varphi) \tilde{\mathbf{S}}_\nu \quad (180)$$

Here $\dot{\tau}_T = n_e \sigma_T a$ is the Thomson scattering rate. In the axisymmetric approximation (azimuthal average), the phase matrix is

$$\mathbf{M}(\mu) = \frac{3}{4} \left\{ (1 + \mu^2, 1 - \mu^2, 0); (1 - \mu^2, 1 + \mu^2, 0); (0, 0, 2\mu) \right\} \quad (181)$$

given by the above expression. The form in which the local quadrupole Π becomes the polarization

source is

$$k^\mu \nabla_\mu^{(g(P))} (\tilde{Q}_\nu \pm i\tilde{U}_\nu) = -\tilde{A}_P (\tilde{Q}_\nu \pm i\tilde{U}_\nu) + \dot{\tau}_T \left[-(\tilde{Q}_\nu \pm i\tilde{U}_\nu) + \frac{3}{4}(1 - \mu^2)\Pi e^{\pm 2i\varphi} \right] \quad (182)$$

fixed as above.

The origin of Π is the fluid-velocity gradient that carries a phase lag of $\pi/2$ relative to the density perturbation. Write the continuity equation as

$$\Theta' + \frac{1}{3} \nabla \cdot v_{pb} = 0, \quad \Theta'_k = -\frac{k}{3} v_{pb,k} \quad (183)$$

and substitute the acoustic mode

$$\Theta_k(r_s) = \Theta_{eq,k} + \Delta\Theta_k \cos(kr_s) \quad (184)$$

to obtain

$$v_{pb,k}(r_s) = 3c_{s,P} \Delta\Theta_k \sin(kr_s), \quad \nabla \cdot v_{pb,k} = ik v_{pb,k} \propto \sin(kr_s) \quad (185)$$

as above. The quadrupole hierarchy closes as

$$\Pi'_k + \dot{\tau}_T \Pi_k = \frac{8}{15} k v_{pb,k} \quad (186)$$

which implies that in the tight-coupling limit ($\dot{\tau}_T \gg \partial_s$)

$$\Pi_k \simeq \frac{8}{15} \frac{k}{\dot{\tau}_T} v_{pb,k} \simeq \frac{8}{15} \frac{1}{\dot{\tau}_T} \nabla \cdot v_{pb,k} \quad (187)$$

so that Π is generated from the velocity gradient. Therefore the polarization source term is proportional to $\dot{\tau}_T(1 - \mu^2)\Pi$, and the E mode is excited on the $\sin(kr_s)$ branch.

For the acoustic phase $x \equiv kr_s$, adopt the minimal approximation

$$\Theta \propto \cos x, \quad E \propto \sin x \quad (188)$$

Then

$$C_\ell^{TT} \propto \cos^2 x, \quad C_\ell^{EE} \propto \sin^2 x, \quad C_\ell^{TE} \propto \sin x \cos x = \frac{1}{2} \sin(2x) \quad (189)$$

the extremum conditions are

$$x_n^{TT} = n\pi, \quad x_n^{EE} = \left(n + \frac{1}{2}\right)\pi, \quad x_m^{TE, \text{ext}} = \left(m + \frac{1}{4}\right)\pi \quad (190)$$

and the phase differences are

$$\Delta x_{EE-TT} = \frac{\pi}{2}, \quad \Delta x_{TE-TT} = \frac{\pi}{4} \quad (191)$$

which are fixed uniquely as the half-wavelength shift and the intermediate phase relation among TT/EE/TE. From this phase relation, using $\ell_n^{TT} \approx (n - \phi)\ell_A$ as the TT reference,

$$\ell_n^{EE} \approx (n - \phi + \frac{1}{2})\ell_A, \quad \ell_m^{TE, \text{ext}} \approx (m - \phi + \frac{1}{4})\ell_A \quad (192)$$

these are fixed as analytic predictions (for TE, the sign alternates from extremum to extremum).

$$\tilde{I}_{\nu, \text{obs}}(b) = \int_{\lambda_{\text{in}}}^{\lambda_{\text{out}}} \tilde{J}_{\nu}(\lambda, b) \exp[-\tau_{\nu}(\lambda, b)] d\lambda, \quad \tau_{\nu}(\lambda, b) = \int_{\lambda}^{\lambda_{\text{out}}} \tilde{A}_{\nu} d\lambda' \quad (193)$$

This is fixed by taking the minimal coupling of P_{μ} and J^{μ} as the primary structure, rather than by replacing it with an already-solved GRMHD background.

The observed intensity is obtained from the redshift factor

$$g_{\text{red}}(b) = \frac{(k_{\mu} u^{\mu})_{\text{obs}}}{(k_{\mu} u^{\mu})_{\text{em}}}, \quad I_{\nu, \text{obs}}(b) = g_{\text{red}}^3(b) \nu_{\text{obs}}^3 \tilde{I}_{\nu, \text{obs}}(b) \quad (194)$$

as above. The ring radius is defined as the solution of $dI_{\nu, \text{obs}}/db = 0$, and

$$\kappa \equiv \frac{b_{\text{peak}}}{b_{\text{sh}}(P)}, \quad \left. \frac{dI_{\nu, \text{obs}}}{db} \right|_{b=b_{\text{peak}}} = 0 \quad (195)$$

is fixed in that way. With the first-order expansion $b_{\text{peak}} = b_{\text{sh}} + \delta b$, one obtains

$$\kappa \simeq 1 + \frac{1}{b_{\text{sh}} \Lambda_2} [\partial_b \ln(g_{\text{red}}^3 \bar{J}_{\nu}) - \partial_b \bar{\tau}_{\nu}]_{b=b_{\text{sh}}}, \quad \Lambda_2 \equiv -\partial_b^2 \ln \mathcal{G}(b)|_{b_{\text{sh}}} > 0 \quad (196)$$

Here $\mathcal{G}(b)$ is the geometric amplification factor near the capture boundary. Therefore κ is not determined by geometry alone, but by (i) $\mathcal{G}$ from $g(P)$, (ii) the radiative gradient from J^{μ} , and (iii) the absorption/scattering contribution entering $\bar{\tau}_{\nu}$.

The analytic boundary conditions are

$$r \rightarrow r_H^+ : k^r < 0, \tilde{I}_{\nu} \text{ finite}, \tau_{\nu} \text{ finite} \quad (197)$$

$$r \rightarrow \infty : u \rightarrow 0, P_{\phi} \rightarrow 0, \tilde{I}_{\nu} \rightarrow 0, \tau_{\nu} \rightarrow 0 \quad (198)$$

$$\theta = 0, \pi : P_{\phi} = 0, \partial_{\theta} \tilde{I}_{\nu} = 0 \quad (199)$$

These are fixed as the analytic boundary conditions. Part II uses this chain of equations to compare the fitted value and the first-principles envelope of κ on the same I/F.[10][11]

2.9 Quantum (Connection Point: Selection)

The connection to quantum physics is a validation target of Part III (Quantum Physics). In this paper (Part I), only the minimal dictionary for the phase mapping and the observation process is fixed, while the detailed acceptance or rejection is delegated to the same audit interface of Part III (input / freezing / statistics / decision / output).

Definition of phase and reduction to the Schrödinger equation: Assume that the phase of a bound mode (particle) accumulates in proportion to the proper time τ ($\psi \propto e^{-i\omega_0\tau}$). Using the clock mapping $d\tau/dt = e^{\phi/c^2} \approx 1 + \phi/c^2$, the phase in the weak field separates as follows.

$$\psi \propto e^{-i\omega_0\tau} \quad (200)$$

$$e^{-i\omega_0\tau} \approx e^{-i\omega_0 t} e^{-i\omega_0 \phi t/c^2} \quad (201)$$

Here, define $E = \hbar\omega_0$ and $m = \hbar\omega_0/c^2$. Applying the nonrelativistic envelope approximation then yields

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + m\phi \right) \psi \quad (202)$$

is recovered. That is, quantum-mechanical behavior in the P-model is positioned as an approximate expression (an effective equation) of deterministic interference generated by fluctuations of time-wave density together with the delay of proper time.

Operational adoption of the Born rule and the limit scale (m_{crit}): This paper does not treat the Born rule (probability density $\propto |\psi|^2$) as a first principle. It is treated as an operational effective law valid in the range where linear response and the Markov approximation hold. The critical mass at which the self-gravity (mean-field) effect of the wave packet becomes manifest is estimated by

$$m_{\text{crit}} \sim \sqrt{\frac{\hbar R}{GT}} \quad (203)$$

This provides the estimate. In addition, wave-packet collapse is described not as something caused by "the observer's consciousness" or by "many worlds," but as a physical time-evolution process driven by the pointer degrees of freedom of the apparatus and by environmental dissipation (branch convergence in a nonlinear continuous-measurement equation). The details are validated in Part III. Likewise, nonlocality in Bell tests and related cases is treated as a systematic effect in which the event-adoption rule (selection) changes the statistical ensemble, and the falsification conditions are constructed within the framework of local realism.[15][16]

3 Methods

3.1 Common Rejection Procedure

Each validation in this paper fixes falsifiability in the following common format.

- **Input:** the data used (primary / already-estimated), dependency assumptions, acquisition source, and local cache name.
- **Freeze:** frozen parameters and the grounds for freezing (which primary constraint fixed them, and when).
- **Output:** fixed filenames, column definitions, units, and reproduction commands.
- **Statistics:** evaluation statistics (for example RMS, χ^2 , logL, ΔAIC , slope). ΔAIC is defined by

$$\Delta\text{AIC} \equiv \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P-model}} \quad (204)$$

as above.

- **Decision:** rejection thresholds (for example exceeding 3σ , $\Delta\text{AIC} < -10$ meaning strong baseline preference, or excess after including systematics).

From this point onward, every test is written in this format so that it is reproducibly rejectable as a protocol, not as a discussion.

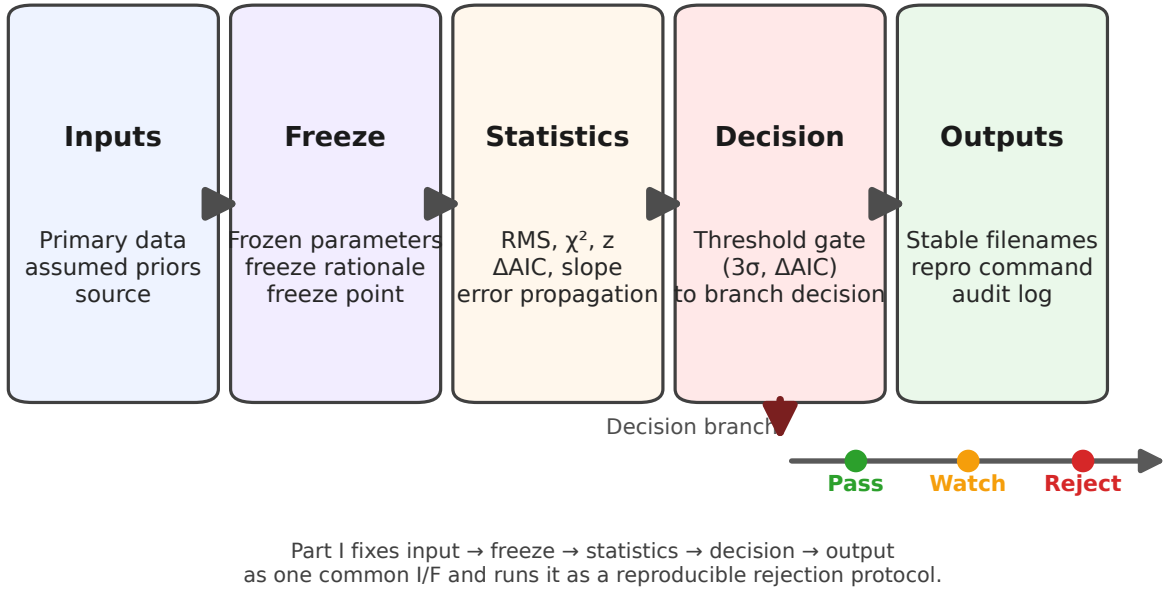


Figure 2: Flowchart of the rejection protocol. It shows the decision chain input → freeze → statistics → decision → output, together with the post-decision branches adopt / watch / reject.

3.2 Reproducibility (Execution and Artifacts)

In this study, the processing conditions that generate the figures, tables, and key numerical values from primary data are fixed for each topic so that the same input yields the same output. However, the paper body itself does not enumerate concrete filenames, folder names, or script names. The information needed for reproduction (processing conditions, primary sources, access dates, frozen parameters, fixed artifact lists, reproduction commands, and audit gates) is governed by Part IV (Verification Materials) (GitHub: [EnterOgawa/p-model](#)). The acceptance of the action principle is given in Part IV, Section 3, and the grounds specific to the Part I mappings are given in Part IV, Section 10.[1]

3.3 Data Acquisition and Local Cache

Data acquired from primary sources are recorded together with their access dates, and the acquired files are managed in a form identifiable by content identity (for example hashes). This makes it possible to trace which data underlie the results of this paper even if the upstream distribution source is later updated.

The list of primary sources is collected in the appendix (Data Sources).

3.4 Three-Layer Definition and Operation of the Common Parameter β

The role of β is managed in three layers: the theoretical prediction, the operational frozen value, and the terminal audit status.

1. **Theoretical prediction:** $\beta = 1$. It is derived from the null geodesics of the effective metric (Section 2.5). It is the reference value that follows from the isotropic form of the P-model ($g_{tt} \cdot g_{rr} = -c^2$).
 2. **Operational frozen value:** $\beta = 1.0000105$. This is the reference value obtained by mapping Cassini's published PPN γ through $\beta = (1 + \gamma)/2$. It is an operational value obtained through the GR-side framework and is not itself the independent validation value of the theoretical prediction. When the difference from $\beta = 1$ is below observational precision in weak-field numerical calculations, this frozen value is used in the extrapolation calculations.[5]
 3. **Terminal audit status:** Pass (current). In the PPN-independent cross-channel audit, both LLR and MESSENGER (beta_lt as the primary estimate) are consistent with $\beta = 1$, and the pair gate has also achieved Pass. The policy governance has already been promoted to policy_D_exclude_plus_messenger_fusion (details are given in Part II 4.4 / the Part IV registry).
- ϕ : the Newtonian-form potential defined from P (Section 2.1). The definition of ϕ is not changed in order to absorb coefficients; if necessary, that is handled on the β side.
 - The frozen values and terminal decisions are stored in the frozen-parameter set (machine-readable), and the same values are referenced in comparison extrapolation and audit.

Reference note: if PPN is used as the reference framework, Cassini's γ constraint can be mapped to β through the correspondence $1 + \gamma = 2\beta$ (the reference-framework limit is $\beta = 1$).[2][5]

Why β is frozen Cassini + VLBI cross-check

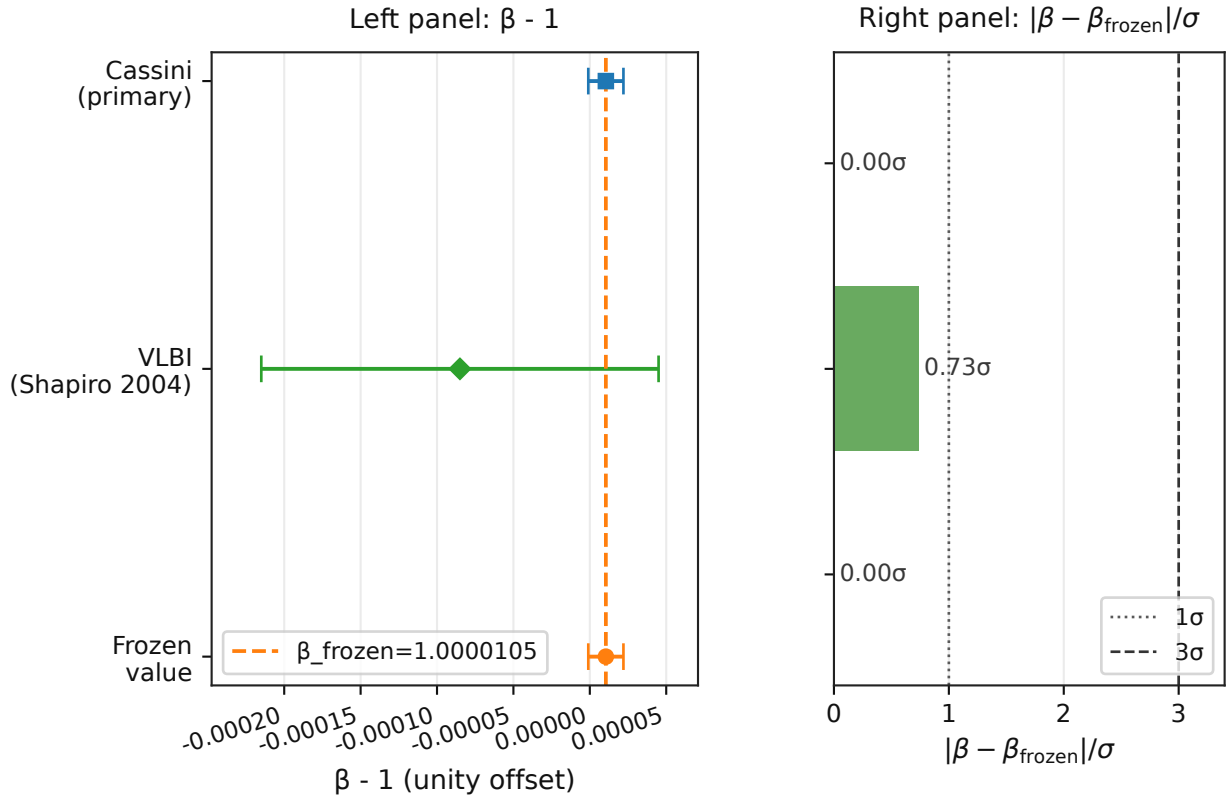


Figure 3: Visualization of the grounds for freezing β . The left panel displays the deviation of $\beta - 1$, while the right panel shows the deviation from the frozen value in units of σ and compares it with the $1\sigma/3\sigma$ thresholds.

4 Summary of Results and Connection to Parts II/III

This paper (Part I) fixes the P-model's **symbol conventions, minimum assumptions, and mappings to observables** as the canonical baseline, summarizes the fixed outputs (falsification conditions and audit gates), and then connects them to Parts II and III. The applied validation against observational data is divided into the following two later parts, both of which reanalyze the data and present falsification conditions under the same conventions fixed in Part I.

The recommended reading order is **Part I -> Part II -> Part III**. Parts II and III assume the vocabulary, sign conventions, and frozen values fixed in this paper and do not rewrite mappings or parameters ad hoc for the convenience of the application sections.

- Part II (Astrophysics)
- Part III (Quantum Physics)
- Part IV (Verification Materials)

4.1 Decision Conditions for the Differential Predictions (Overview)

The condition under which the P-model becomes observationally distinguishable from the reference frameworks is that the **differential predictions exceed the statistical and systematic uncertainties**. This paper (Part I) leaves the detailed calculations and primary-source organization to Parts II and III, while fixing only the main decision points, that is, which classes of observations can settle the issue.

1. Terminal status of the free-wave propagation coefficient β (LLR / MESSENGER / VLBI eligibility chain)

- In the P-model, the light-response exponent β is theoretically predicted to be $\beta = 1$ from the null geodesics of the effective metric.
- If β is not exactly 1, systematic residual patterns appear in light bending and Shapiro delay.
- Decision condition: whether LLR and MESSENGER (through β_{tt}) satisfy consistency with the theoretical prediction $\beta = 1$. VLBI remains an auxiliary audit until the eligibility of the comparison baseline is established.

2. Shape observables in the strong-gravity regime (nonlinear structure of the P gradient)

- In the P-model, the structure of the P gradient in the strong-field regime may differ from that of the reference framework.
- The difference appears as coefficient shifts in shape observables such as the shadow diameter and ring morphology.
- Decision condition: whether EHT-class shape observables become statistically distinguishable from the theoretical prediction.

3. Cosmological distance-indicator-independent quantities (validation of the background-P mapping)

- In the P-model, redshift is explained without spatial expansion, through the time evolution of the background P field.
- It can therefore be tested with observables that do not depend on distance indicators, such as time-dilation scaling and the CMB temperature.
- Decision condition: whether distance-independent probes are consistent with the background-P mapping.

4. Simultaneous success of rotation and polarization (survival condition of the P_μ extension)

- The reason for introducing the P_μ extension is that it permits a simultaneous description of rotating sources (frame dragging) and gravitational-wave polarization.
- Decision condition: simultaneously achieving scalar-polarization rejection and tensor-polarization consistency with a three-detector network (H1/L1/V1).[12]
- This simultaneous achievement is the highest-priority audit gate on the Part II side; if it is not met, the effectiveness of the P_μ extension remains under continuing audit (watch).

5. Resolution of nonlocality on the quantum scale and the microscopic coupling of P_μ

- The P-model treats the nonlocality of quantum entanglement as a statistical-ensemble variation (a systematic effect) induced by the event-adoption rule (selection).
- For the microscopic nuclear force (tensor force) and the V-A structure of the weak interaction, the candidate dictionary connecting P_μ to spin and chiral currents is retained as the reference note of Part I 2.8.2.
- Decision condition: Bell tests must be carried out with a setting-independent ensemble. For the weak-interaction candidate dictionary, a full tensor calculation using P_μ must reproduce observables such as Q values and half-lives.
- At present, however, the authoritative decision remains the Part III-B 5.5 outcome of A adopted / B rejected. Route B can be reopened only if the full tensor calculation above demonstrates reproducibility and triggers re-audit.

6. Frozen-action reference skeleton and the theoretical anchor points on the quantum side

- From the frozen-action reference skeleton fixed in this paper, the transverse component of P_μ yields the transverse photon branch and the massless photon mode (the canonical derivation is Part III-A 2.6).
- The same frozen-action skeleton provides the starting point for the derivation of the fine-structure constant described in detail in Part III-A. Part I limits itself to defining that starting-point action, while the numerical endpoint and residual evaluation are delegated to Part III-A.
- Likewise, the coupled-localization condition for the W/Z boson masses in the two-component coupled Q-ball construction is obtained from the same frozen-action skeleton. The canonical theoretical anchor point and the observational acceptance or rejection are governed by Parts III-A/B.

- Decision condition: Part I fixes only the starting point that provides the quantum-side theoretical anchor points; the detailed derivations and the observational acceptance or rejection are delegated to the audit framework of Parts III-A/B.

The decision points for the differential predictions are fixed in the correspondence between observable, required precision, and rejection condition.

Decision point	Representative observable	Required precision (guide)	Fixed rejection condition
Terminal status of the free-wave propagation coefficient β (LLR, MESSENGER, VLBI eligibility chain)	LLR + MESSENGER (β_{lt}) + the audit of VLBI baseline eligibility. Acceptance or rejection is decided against the theoretical prediction $\beta = 1$ under the policy_D governance rule.	Precision sufficient to decide the LLR/MESSENGER pair gate	Reject if the pair gate fails, or if the terminal status cannot be fixed to Pass because the governance conditions are not satisfied
Shape observables in the strong-gravity regime	EHT ring-shape coefficients (κ, ϵ)	Shape systematic errors reduced to ≤ 0.2 of the current level	Reject if high-precision data continue to favor GR over the P-model (for example, persistent $\Delta AIC \leq 2$ against the P-model)
Distance-independent cosmological quantities	Time and temperature scalings tied to the background mapping	Precision high enough to audit multiple distance-independent probes under the same mapping	Reject if multiple probes continue to show inconsistency larger than 3σ
Simultaneous success of rotation and polarization	Scalar-polarization rejection and tensor-polarization consistency in GW observations	Securing usable events with three or more detectors	Reject if scalar-polarization rejection and tensor-polarization consistency cannot be achieved simultaneously
Quantum scale (selection and microscopic coupling)	Bell fixed-window CHSH/CH tests and weak-interaction Q values / half-lives	Statistical decisions under pre-fixed windows and numerical verification of the full tensor route	Reject if excess correlations continue to be reproduced under fixed windows, or if the microscopic coupling fails to reproduce the observables
Electromagnetic constants and weak-boson masses	Selector-law anchor of the fine-structure constant and the W/Z mass anchor point under the coupled-localization condition	Precision sufficient to reduce the residual of the fine-structure constant and keep $\kappa_{\text{coupled}} > 0$ at the W/Z mass anchor point	Reject if the frozen-action skeleton fails to provide the selector-law anchor of the fine-structure constant, or if the W/Z mass anchor point collapses under the coupled-localization condition

The concrete observables, primary sources, and error budgets (required precision) are collected in Parts II and III.

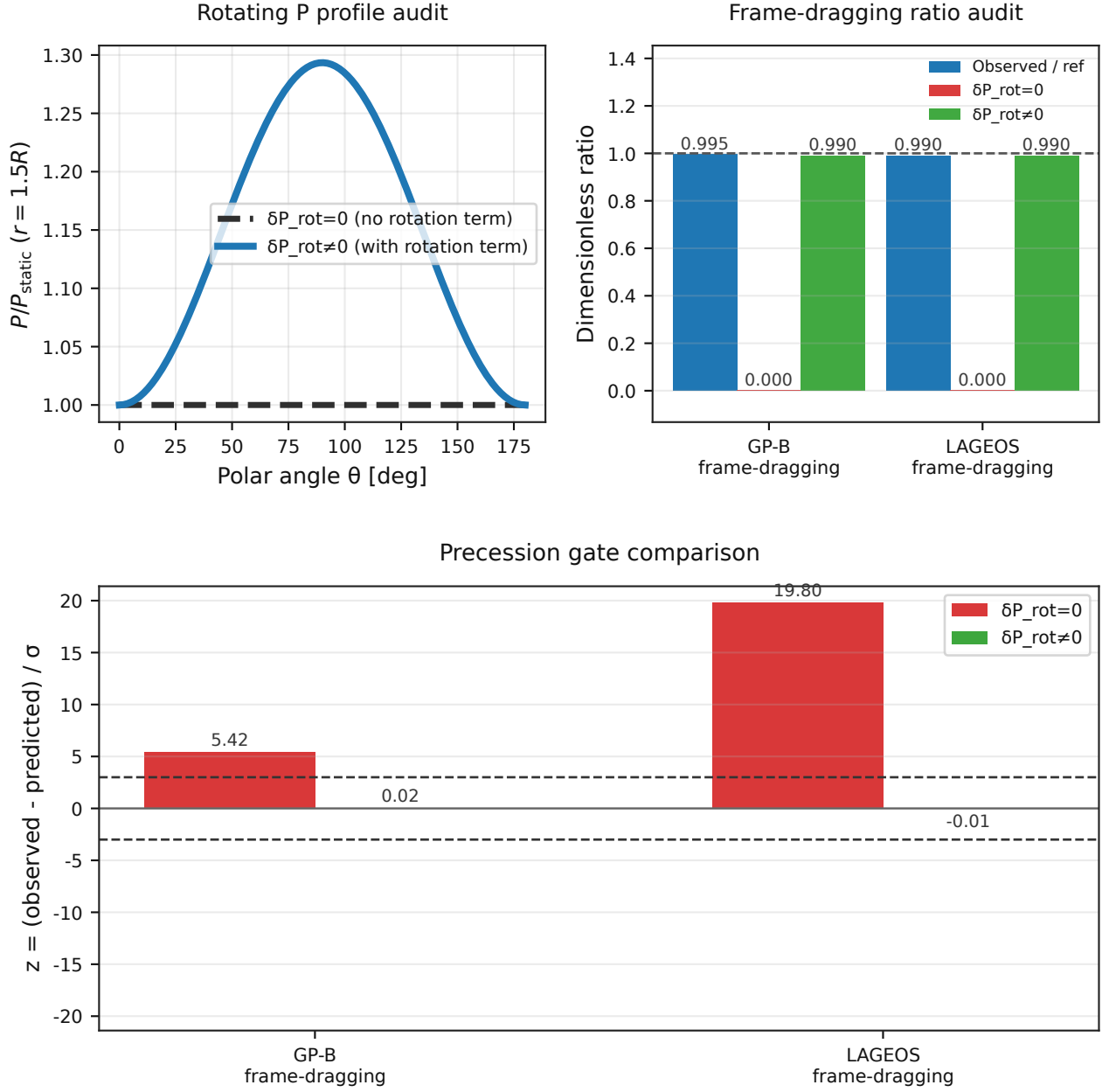


Figure 4: Audit of the P/P_∞ distribution in the rotating-sphere model. It compares the radial and angular distributions together with the normalized stationary-branch ratio at the reference radius $r = 1.5R$, thereby visualizing the transition from the weak to the strong field.

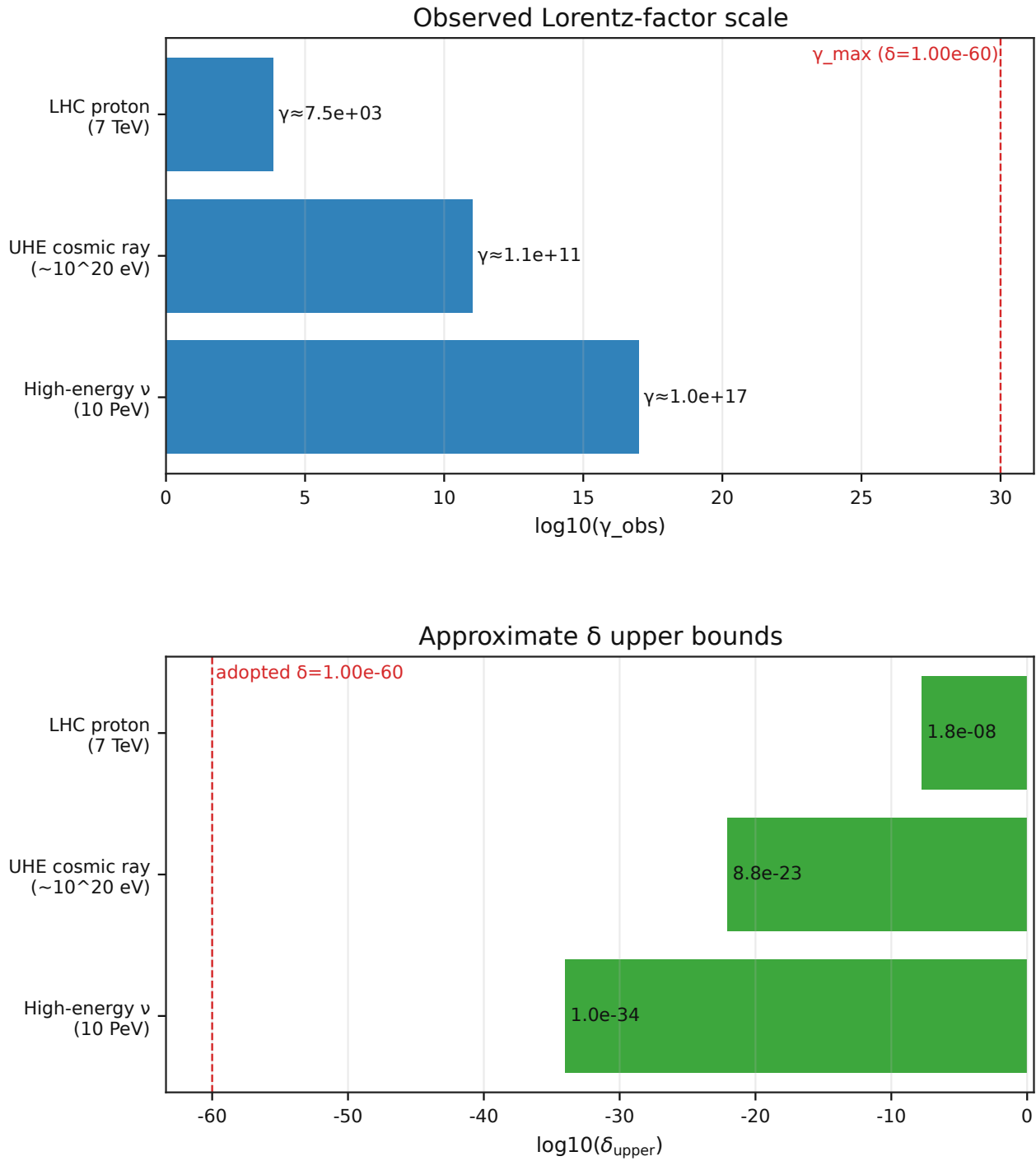


Figure 5: Constraint plot for the velocity-saturation parameter δ . It places upper bounds inferred from the Lorentz factors γ reached by existing observations side by side and compares the order of the allowed range.

P_μ minimal extension: dipole-cancellation audit

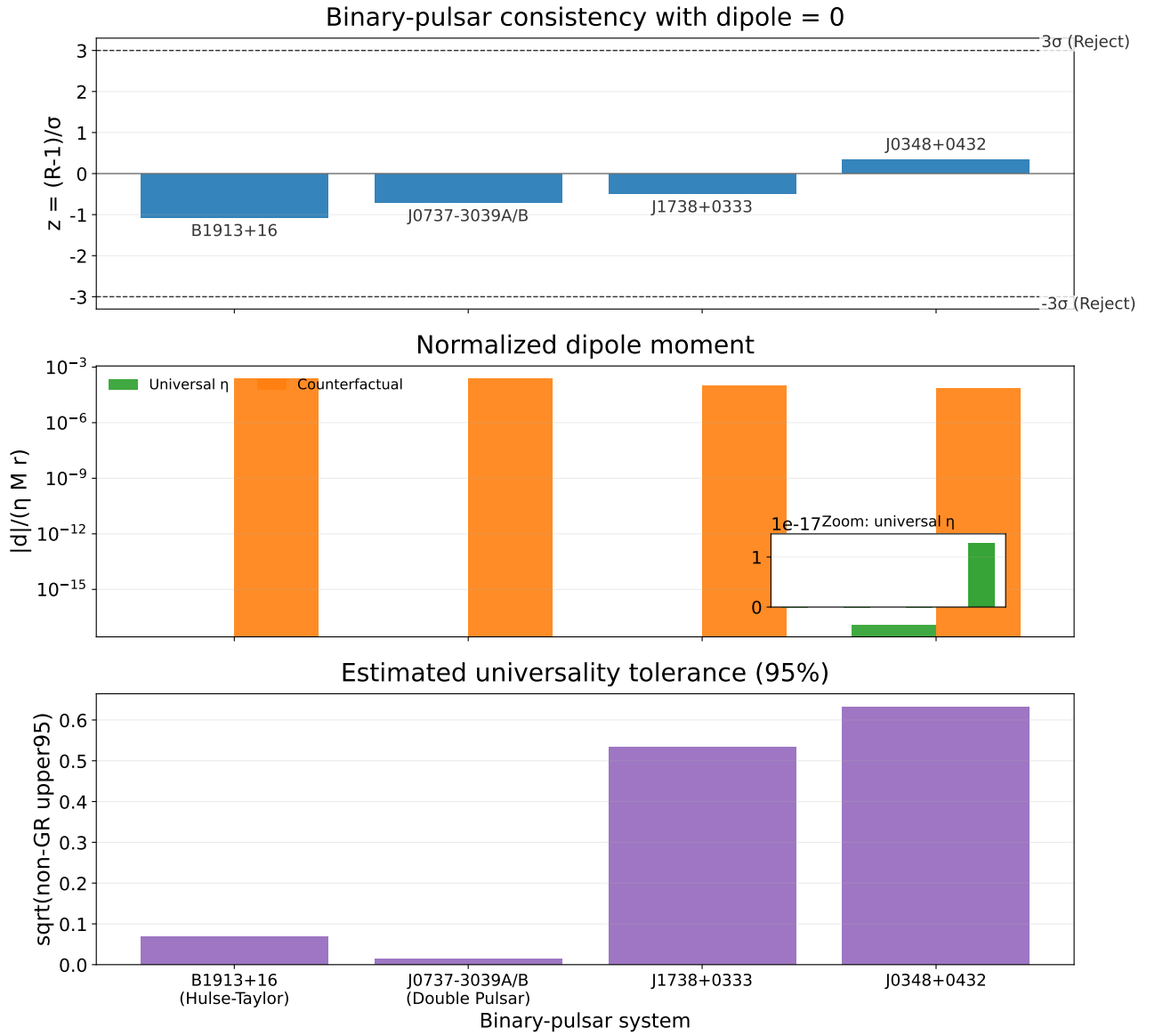


Figure 6: Audit of vector dipole cancellation. The upper panel shows the z-test decision, the middle panel compares the universal and counterfactual branches and enlarges the universal branch, and the lower panel visualizes the allowed widths for each system.

5 Discussion

5.1 What This Paper (Part I) Has Fixed

- Fixed mappings: $\phi = -c^2 \ln(P/P_\infty)$, $\mathbf{a} = c^2 \nabla \ln(P/P_\infty)$, $\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt}\right)_v$, and $n(P) = (P/P_\infty)^{2\beta}$ have been established as the unified conventions.
- Minimal closure of the dynamical branch: the formula chain for the retarded potential, dipole-radiation prohibition, and coarse-grained relaxation (τ, ξ) has been fixed.
- Introduction of the rotational extension: the minimal coupling between P_μ and J^μ , the Stückelberg closure, and the entry point to the strong-field equations have been made explicit and connected to the audit interfaces of Parts II and III.

5.2 Explicit Limitations of This Paper Taken Alone

- The quantum connection remains at the operational stage, and the final acceptance of nonlocal correlations depends on the primary-data audit of Part III.
- While the theoretical prediction $\beta = 1$ has already been derived from the null geodesics of the effective metric, its microscopic origin (why the isotropic effective metric holds) has not yet been identified.
- The final decisions in the strong-field regime (EHT/GW) and for cosmological first-order statistics (BAO/CMB/RSD) are reached through the statistical and systematic gates on the Part II side.

6 Conclusion

This paper (Part I) fixed the symbol conventions, minimum assumptions, mappings to observables, and falsification procedures as the reference baseline of the P-model. The key point is that it placed the stationary-limit scalar branch and the P_μ extension including rotation under the same reference skeleton and thereby supplied the foundation on which the later validations (Parts II and III) become reproducible and rejectable under the same interface.

In addition, the same frozen-action reference skeleton provides the quantum-side theoretical anchor points audited in Parts III-A/B: the transverse photon branch and selector-law anchor of the fine-structure constant, together with the coupled-localization condition for the W/Z boson masses in the two-component coupled Q-ball picture. The derivations and the observational acceptance or rejection of these items are governed by Parts III-A/B.

At the same time, this paper made explicit both the scope and the limits of the theory. The final decisions on the quantum side and in cosmological strong-field regimes are delegated to Parts II and III, while the responsibility of Part I alone is to fix which equations are taken as baseline and what is adopted as the falsification conditions. This separation prevents the claims of the theory from being confused with the claims of its validation and preserves a consistent auditability across the whole series.

7 Appendix A. Symbols and Definitions (Minimal Table)

Symbol	Definition	Note
P	stationary-limit time-wave density	the basic quantity appearing in observations is the ratio P/P_∞
P_∞	far-field reference time-wave density	constant in local systems
P_{ref}	reference time-wave density used in strong-field normalization	for example $u = \ln(P_t/P_{\text{ref}})$
P_μ	four-vector time wave (time component P_t and spatial components P_i)	used in the rotation and flow channels
ϕ	$-c^2 \ln(P/P_\infty)$	zero at infinity and negative near mass
β	light-propagation response index	the frozen value is given in Part I 3.4
u	$\ln(P/P_\infty)$	basic variable of the dynamical equation
D	$c_w^2 \tau_{\text{free}}$	diffusion coefficient in the dynamical branch (2.6)
G_δ	$\delta/\delta_{\text{ref}}$	growth normalization factor in structure formation (2.6.1)
τ_{free}	L_{path}/c_w	free-propagation timescale of the coarse-grained channel
τ_{eff}	$\tau_{\text{free}} + \eta \tau_{\text{kernel}}$	effective timescale of delayed response
ξ	$(\tau_{\text{int}} + \tau_{\text{damp}})/\tau_{\text{eff}}$	strength coefficient of the coarse-grained coupling
κ	$b_{\text{peak}}/b_{\text{sh}}(P)$	EHT ring/shadow mapping coefficient
q_B	radiation-dominated exponent ($q_B = 1/2$)	fixes the asymptotic law of the background wave u_{bg}
q_m	matter-dominated exponent ($q_m = 2/3$)	$P_{\text{bg}}^{(m)} \propto t^{-2/3}$
C_0	integration constant in $\dot{u}_{\text{bg}}^2 - \frac{\Omega_B^2}{2} e^{4(u_{\text{bg}} - u_B)} = C_0$	determines the branching of the exact solution family
Ω_r^2	$8\pi G \rho_{r,0}$	radiation-source coefficient of the background-wave equation
Ω_m^2	$4\pi G \rho_{m,0}$	matter-source coefficient of the background-wave equation

Continued on next page

Symbol	Definition	Note
Ω_Λ^2	$8\pi G\rho_{\Lambda,0}$	present-era term in the background-wave equation (contributing with a negative sign)
g_P	minimal coupling coefficient between P_μ and the matter current J^μ	interaction term $\mathcal{L}_{\text{int}} = g_P P_\mu J^\mu$
Z_P	wavefunction-normalization coefficient of the vector-field kinetic term	$-\frac{Z_P}{4} F_{\mu\nu}^{(P)} F_{(P)}^{\mu\nu}$
m_P	effective mass scale of P_μ	used in the Stückelberg closure
ξ_g	gauge-fixing coefficient	$-\frac{1}{2\xi_g} (\partial_\mu P^\mu + \xi_g m_P \pi)^2$
ξ_{rot}	response coefficient of the rotation channel	sets the strength of the frame-dragging mapping
λ_{rot}	dimensionless coefficient of the rotation coupling	connects macroscopic rotation (Part II) to microscopic interaction (Part III)
μ_{drag}	rotating-source coefficient $g_{\{P\} J_* / (2c^2)$	coefficient in $P_\phi = (\mu_{\text{drag}} \sin^2 \theta) / r^2$
$\mathcal{R}_{\text{drag}}$	observed frame-dragging ratio $\Omega_{\text{obs}} / \Omega_{\text{pred}}$	in the GR reference one has $\mathcal{R}_{\text{drag}} = 1$
J^μ	matter current	enters the minimal coupling term $g_P P_\mu J^\mu$

8 Appendix B. Data Sources and Reproducibility Entry (Minimal)

The canonical registry for primary-source URLs, access dates, fixed artifact lists, reproduction commands, and audit gates is collected in Part IV (Verification Materials). In the Part I main text, only the reproducibility path is fixed; the actual file layout and operation logs are managed mechanically in the Part IV registry.[1]

The representative primary sources used repeatedly in this paper are, for light propagation, Cassini / LLR / MESSENGER (with VLBI as an auxiliary audit)[5][4]; for clocks, GP-A / Galileo[6][7]; for rotation, GP-B / LAGEOS[8][9]; for strong gravity, EHT[10][11]; for gravitational waves, LIGO[12]; for cosmology, Planck / BOSS[13][14]; and for quantum correlations, Bell[15][16].

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